Airframe Technology Master

Design
Lecture 27
Flight Controls Systems I

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- System installation requirements
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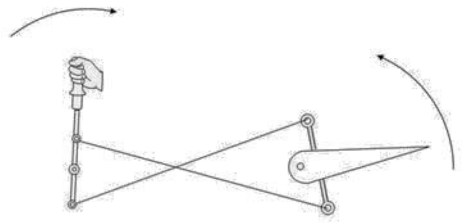
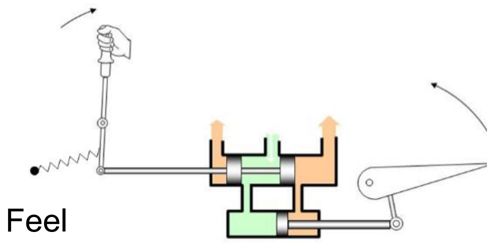
Aircraft Controls definition

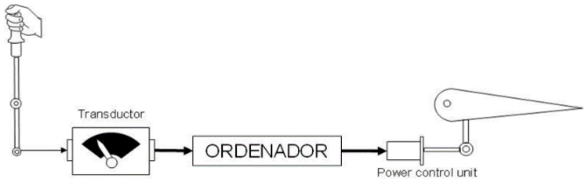
- Aircraft controls are those devices used to transmit the inputs from the pilots, or another automatic control, to the systems to be actuated
- They can be
  - Mechanical (manual): devices are actuated by the action of the pilots directly



  - Assisted (hydraulic, electrical): the control of the mechanical devices is commanded by the pilots directly but the final movement is done by actuators



  - Fly-by-wire: inputs from pilots are processed by several sensors and electrically transmitted to the actuators which operate the control surface or the mechanical device



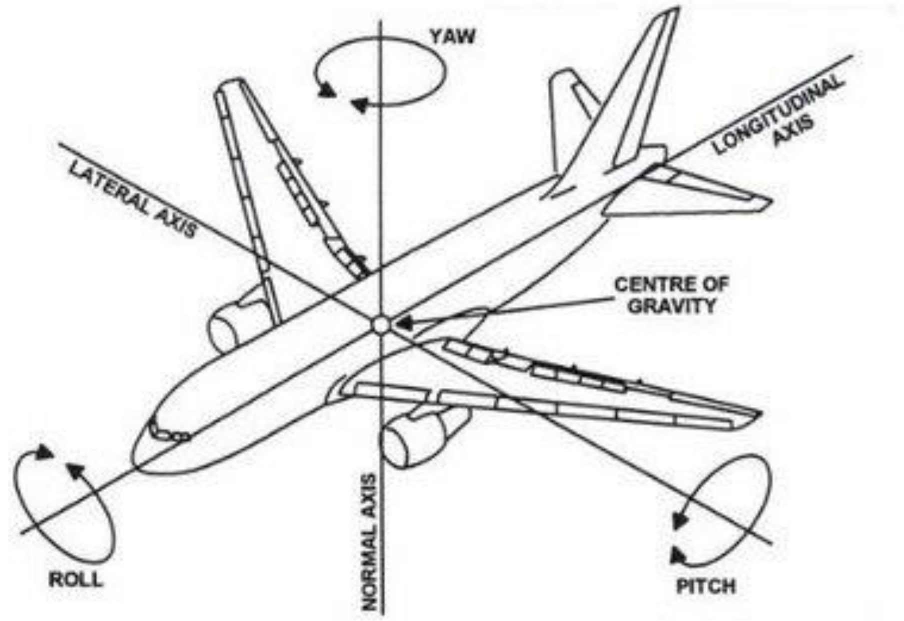
Aircraft Controls types

- Flight controls
  - Primary
  - Secondary
  - Trim´s
  - Autopilot
  - Stick pusher
- Engine controls
- Landing gear controls
  - Landing gear installation
  - Covers system
  - Brake control system
  - Emergency landing gear system
  - Steering control
- Miscellaneous
  - Gust lock
  - Mission systems
  - Protection covers
  - Service systems
  - Doors/cargo doors

General view

— Definitions

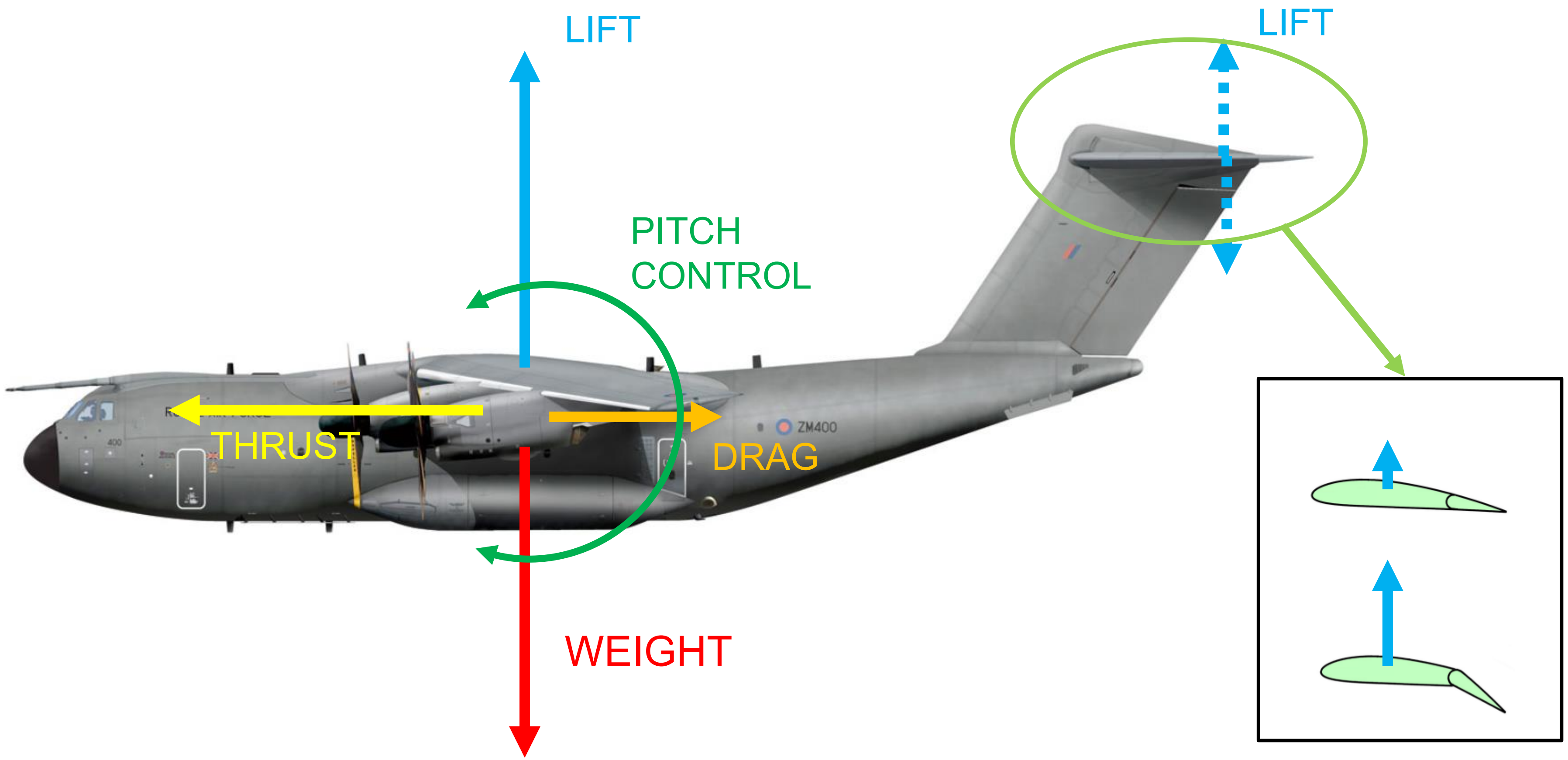
Types of stability and motion



Stability	Axes	Motion about the Axis
Longitudinal	Lateral	Pitch
Lateral	Longitudinal	Roll
Directional	Normal	Yaw

General view

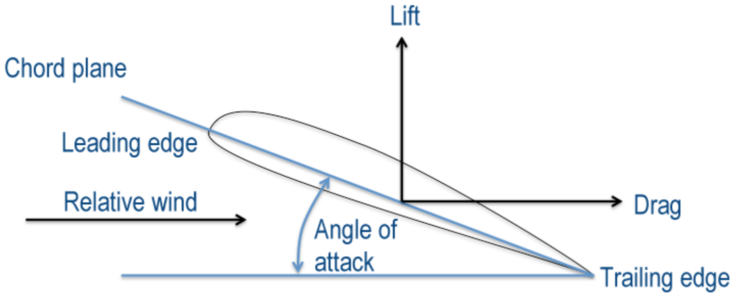
— Definitions: Forces distribution

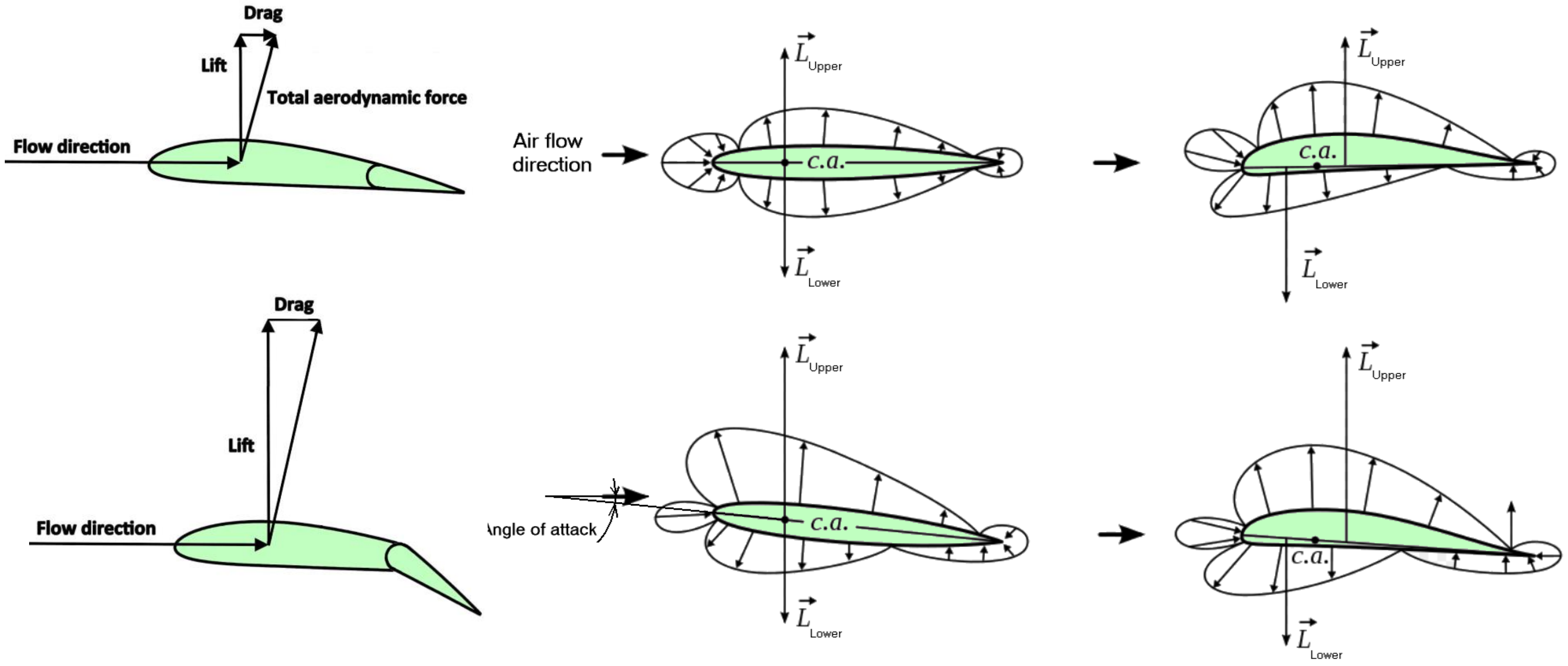


Aircraft flight control surfaces

General view

— Definitions: Aerodynamic forces



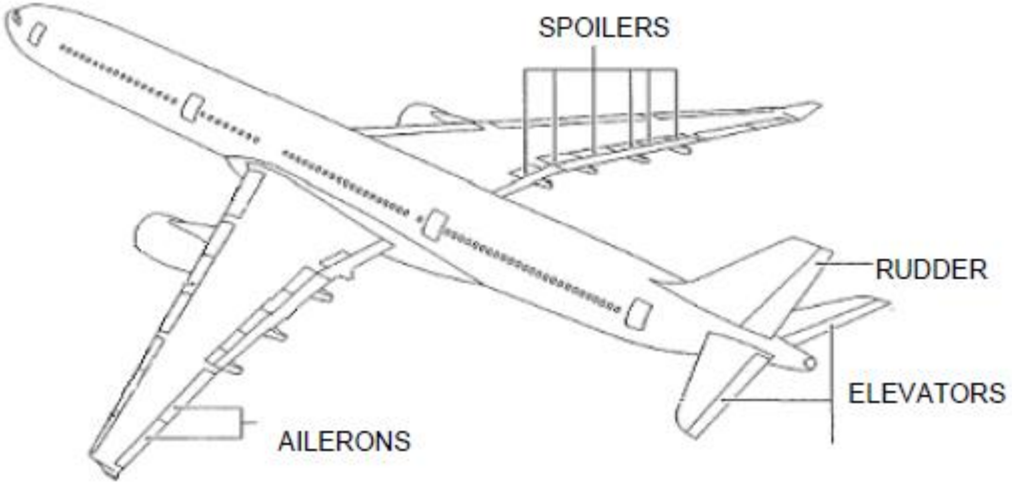


Symmetrical profile

Asymmetrical profile

Aircraft Flight Controls types

— Primary controls are those which are used to maneuver the aircraft in flight

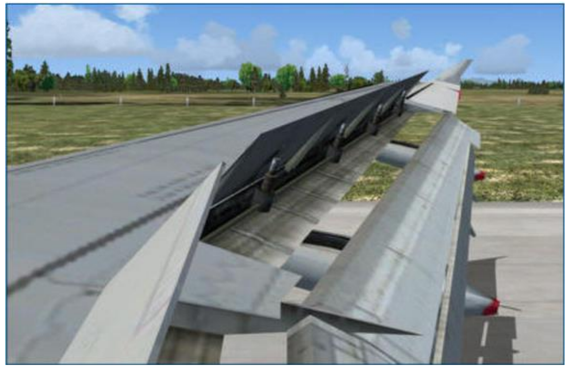


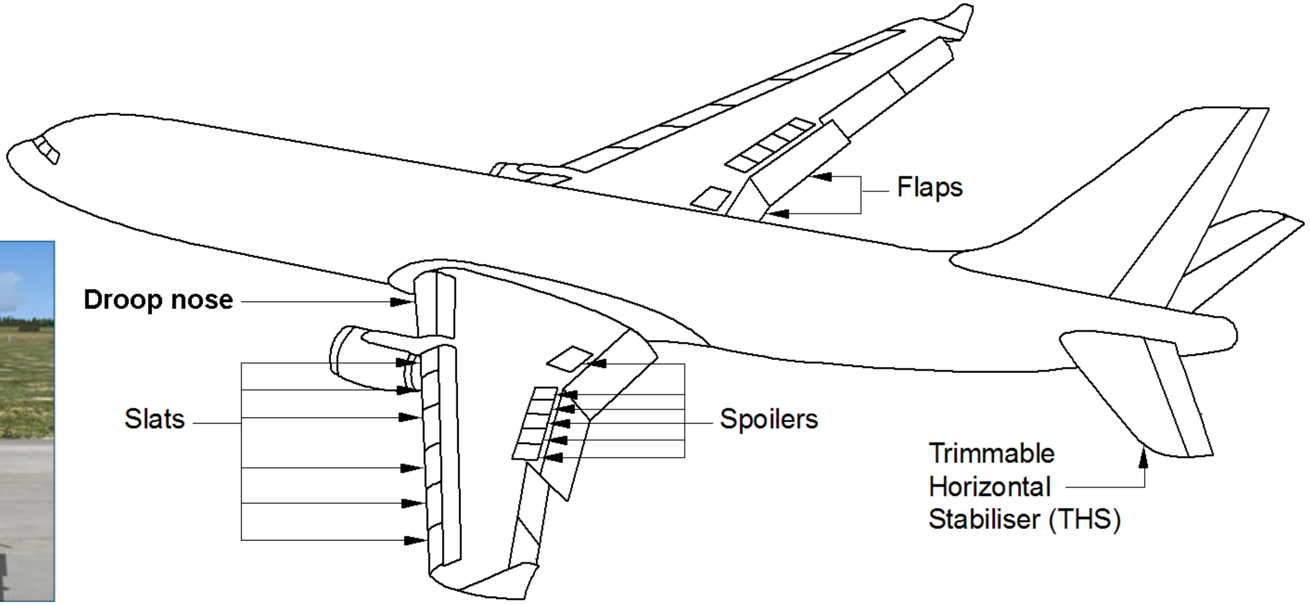
- Elevators: are used to control the pitch aircraft attitude
- Rudder: its deflection causes a yaw motion of the airplane
- Ailerons: their asymmetric deflection cause a roll movement of the aircraft
- Spoilers: are used to help generating rolling movement

— This controls are operated during all fly phases constantly; surfaces can be located in any position of the functional range

Aircraft Flight Controls types

- Secondary controls (or high lift devices) are those which are used to increase the lift capacity of the wings during take-off, approach or landing stages
  - Flap
  - Slat

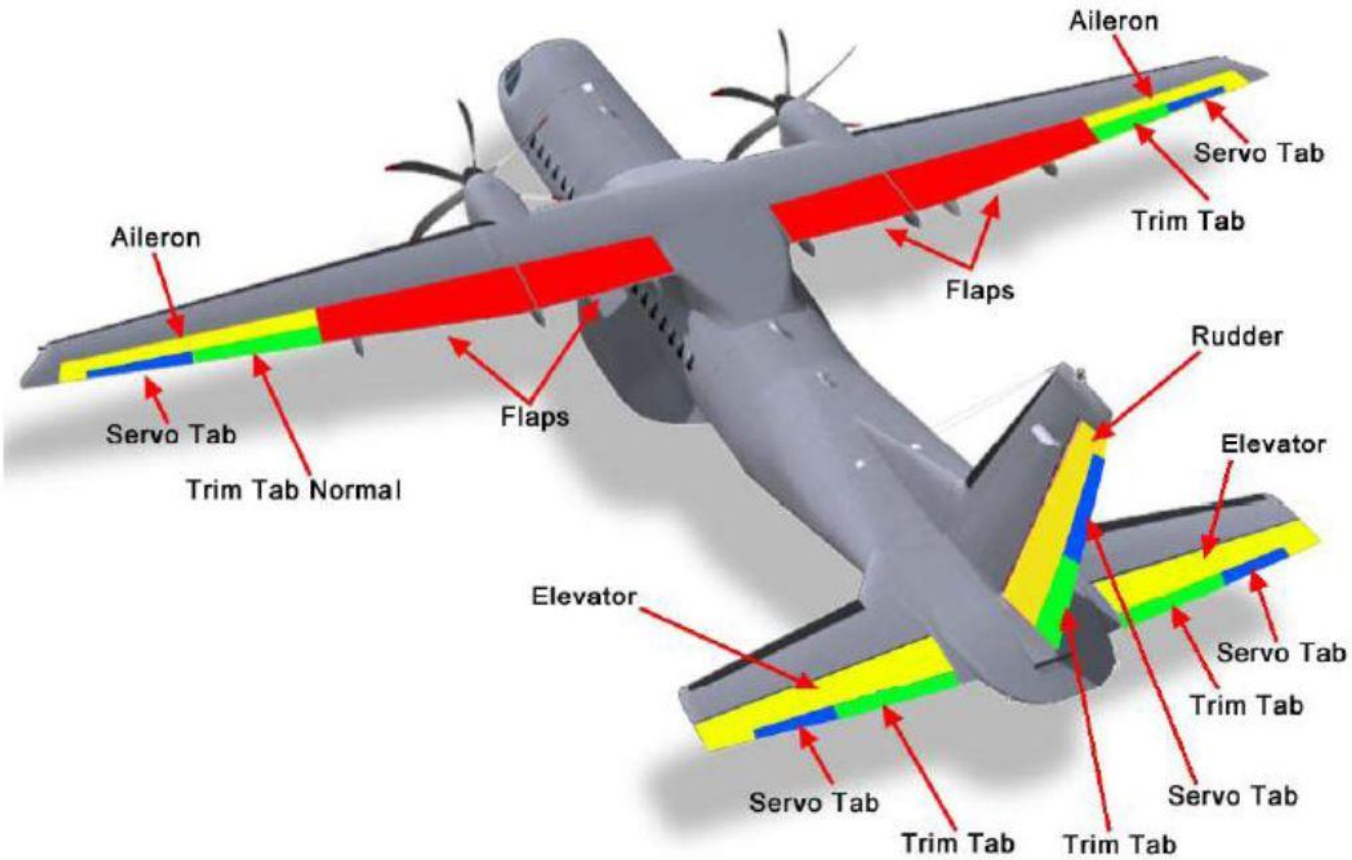




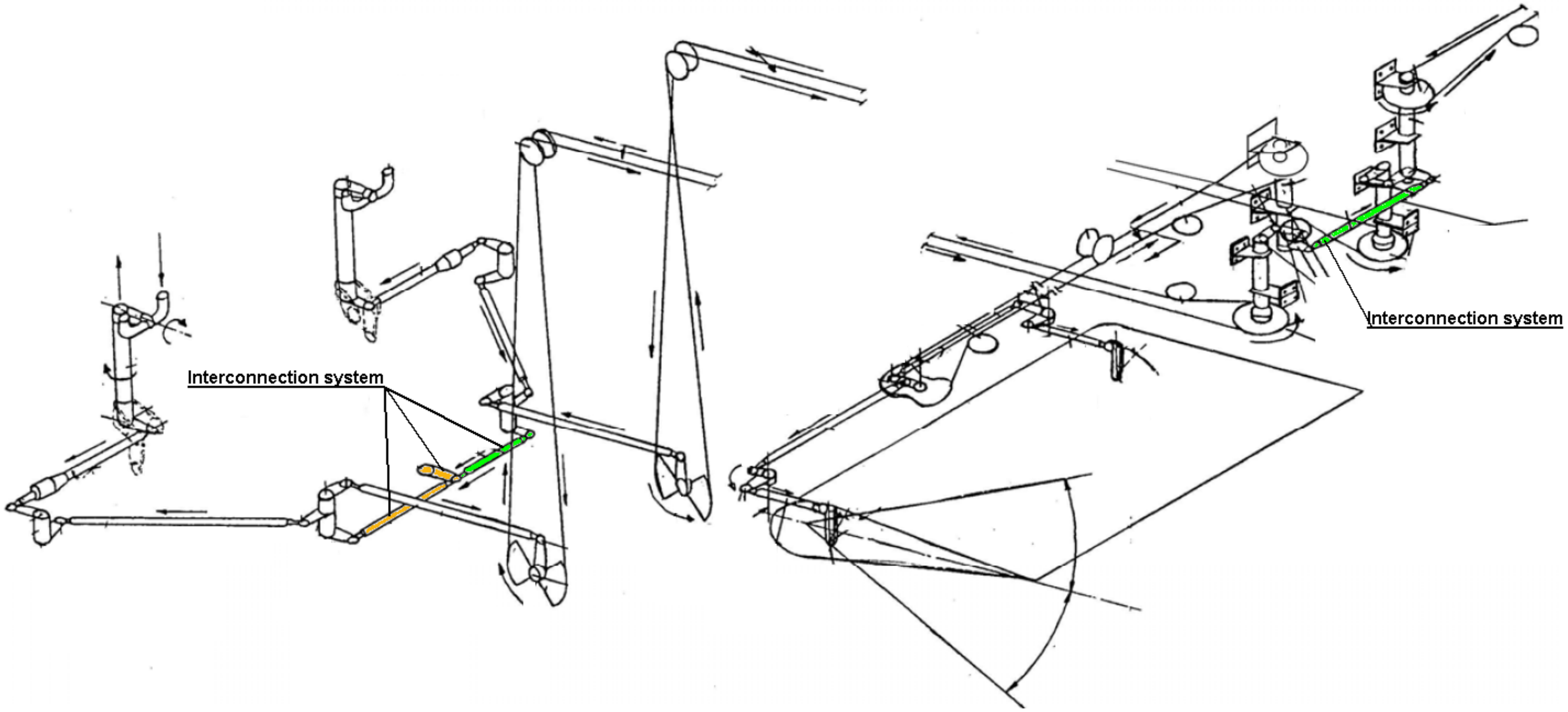
- Trims: are used to modify the angle of attack or the curvature of the surfaces
- This controls are used during certain fly phases modifying wing configuration; surfaces are moved to fixed positions inside the functional range
- Spoilers: used on ground to remove the lift of the wings

Aircraft Flight Controls types

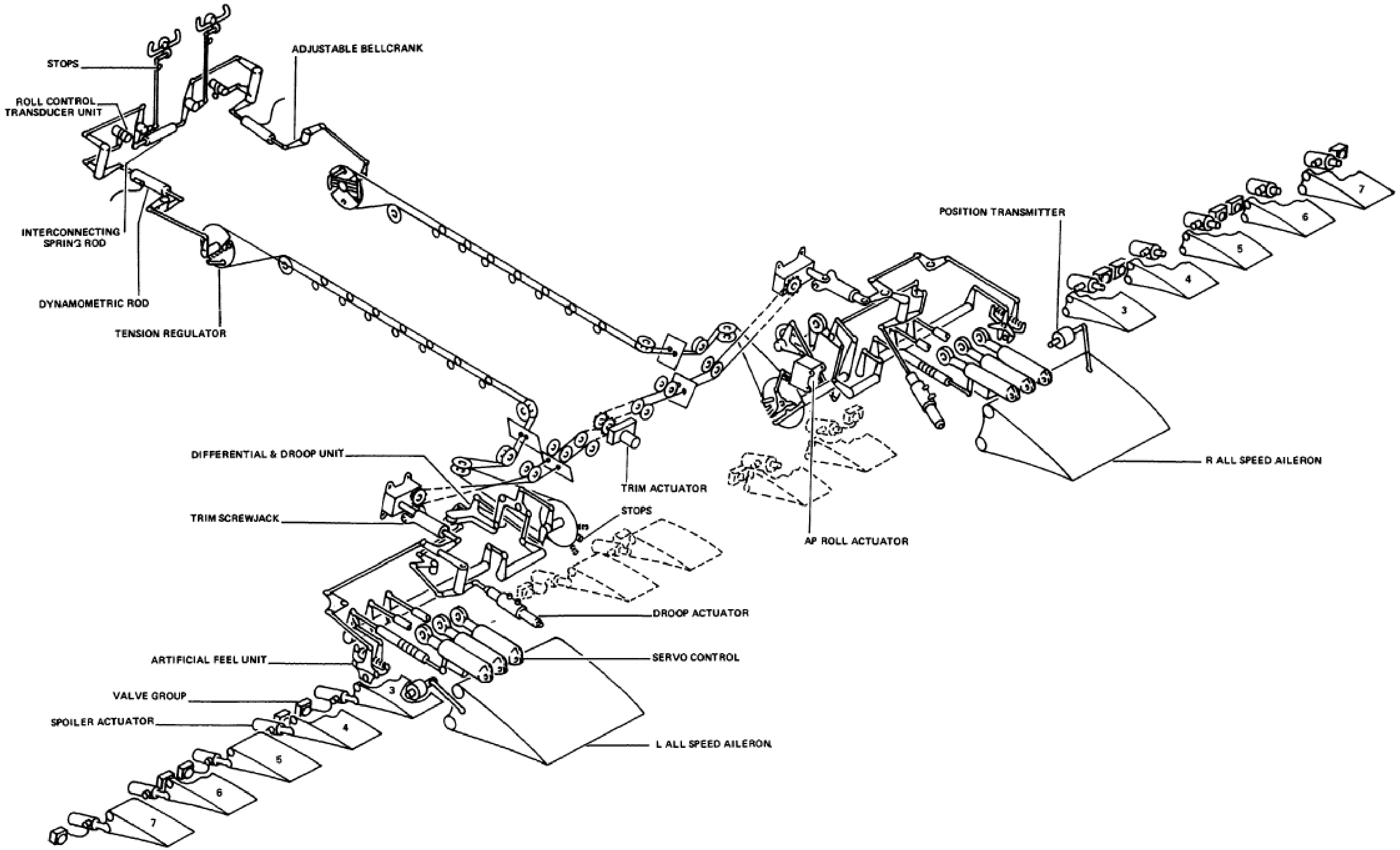
- Mechanical controls: primary/secondary controls, trim´s, servo tabs



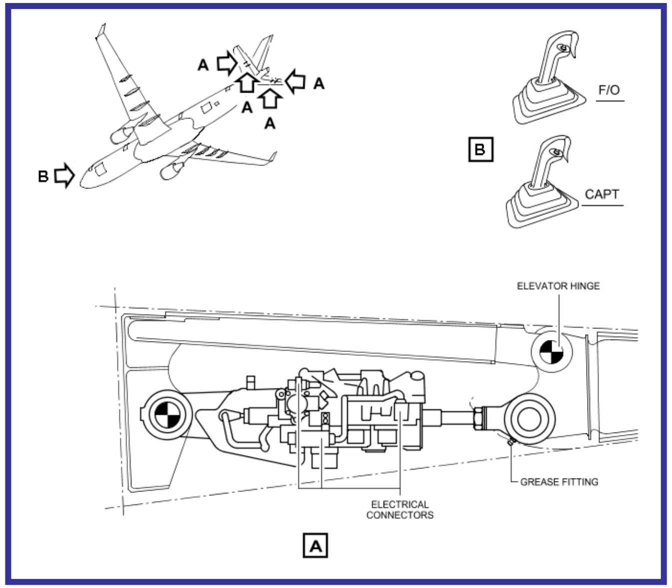
Mechanical controls – roll control - manual

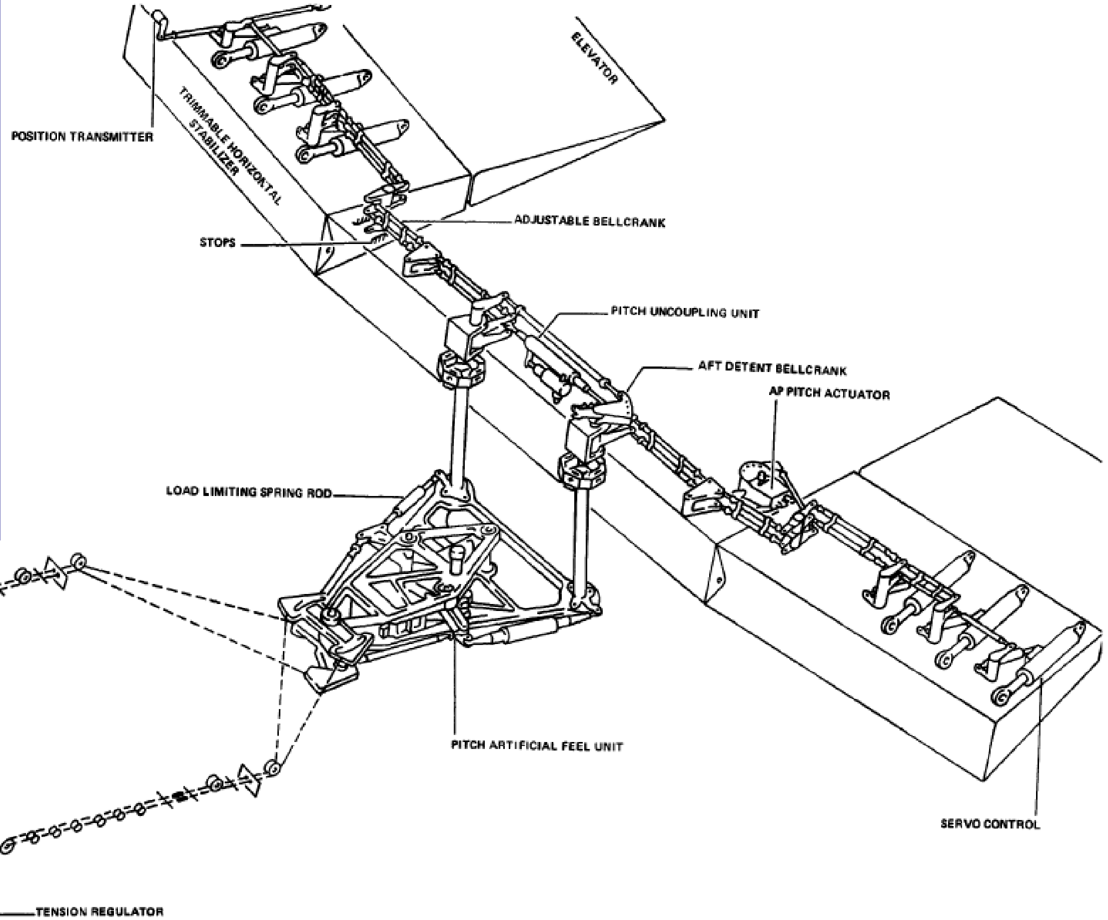


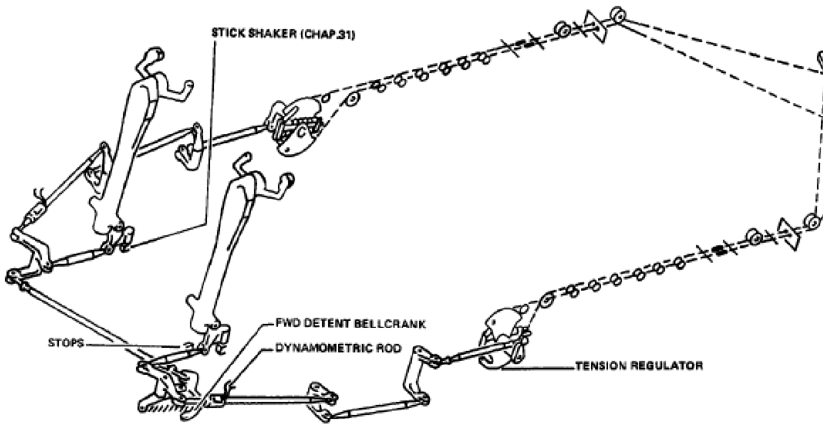
Mechanical controls – roll control - assisted



Mechanical controls – pitch control – assisted vs fly-by-wire

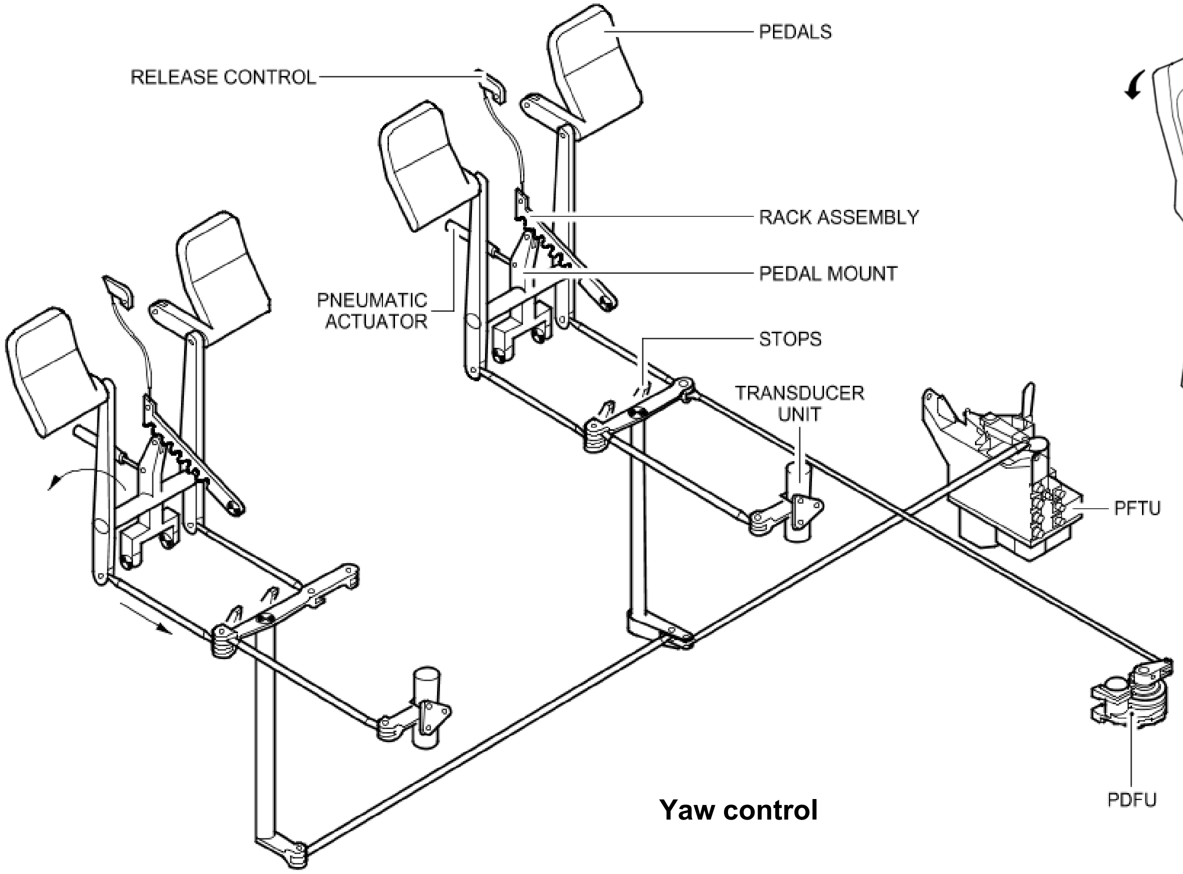




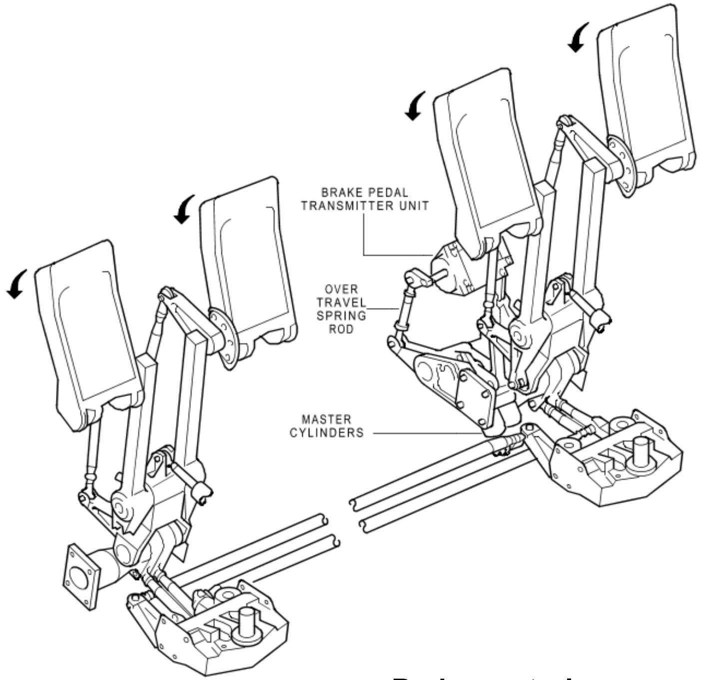


Elevator Mechanical Control

Fly by wire controls – pedals - yaw & brake control

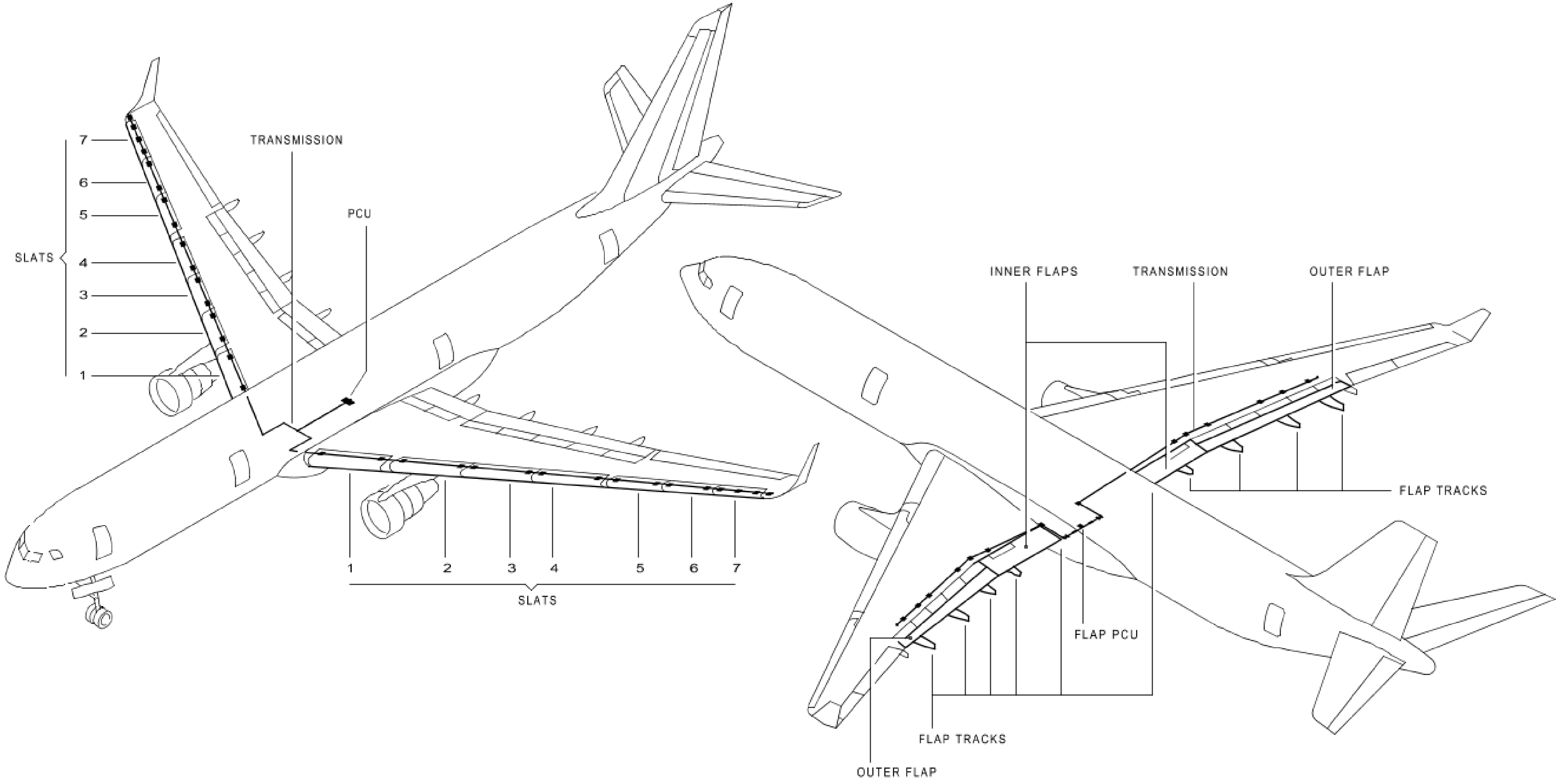


Yaw control

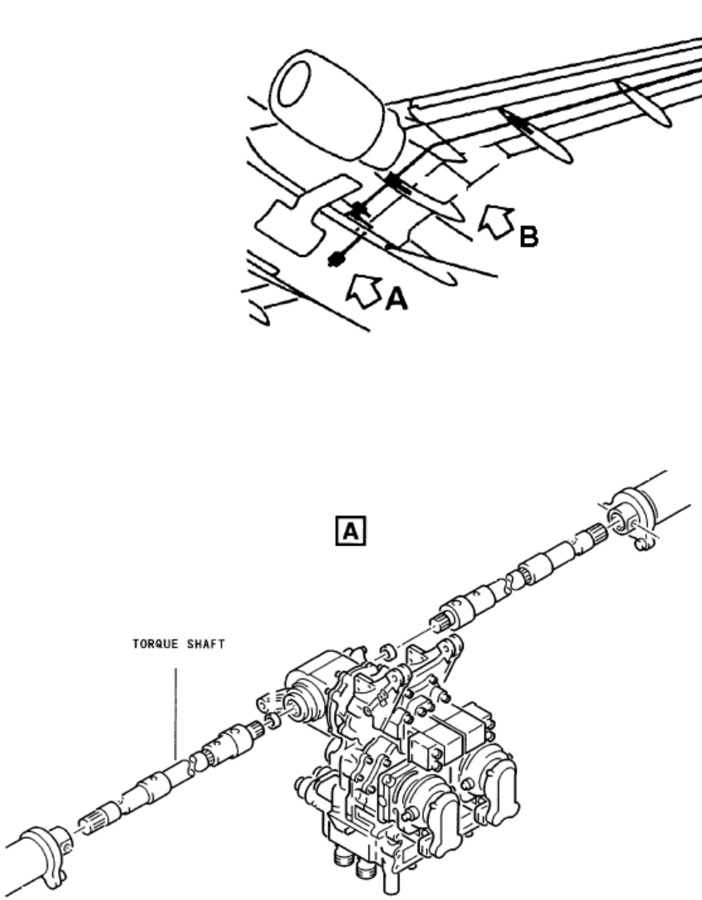


Brake control

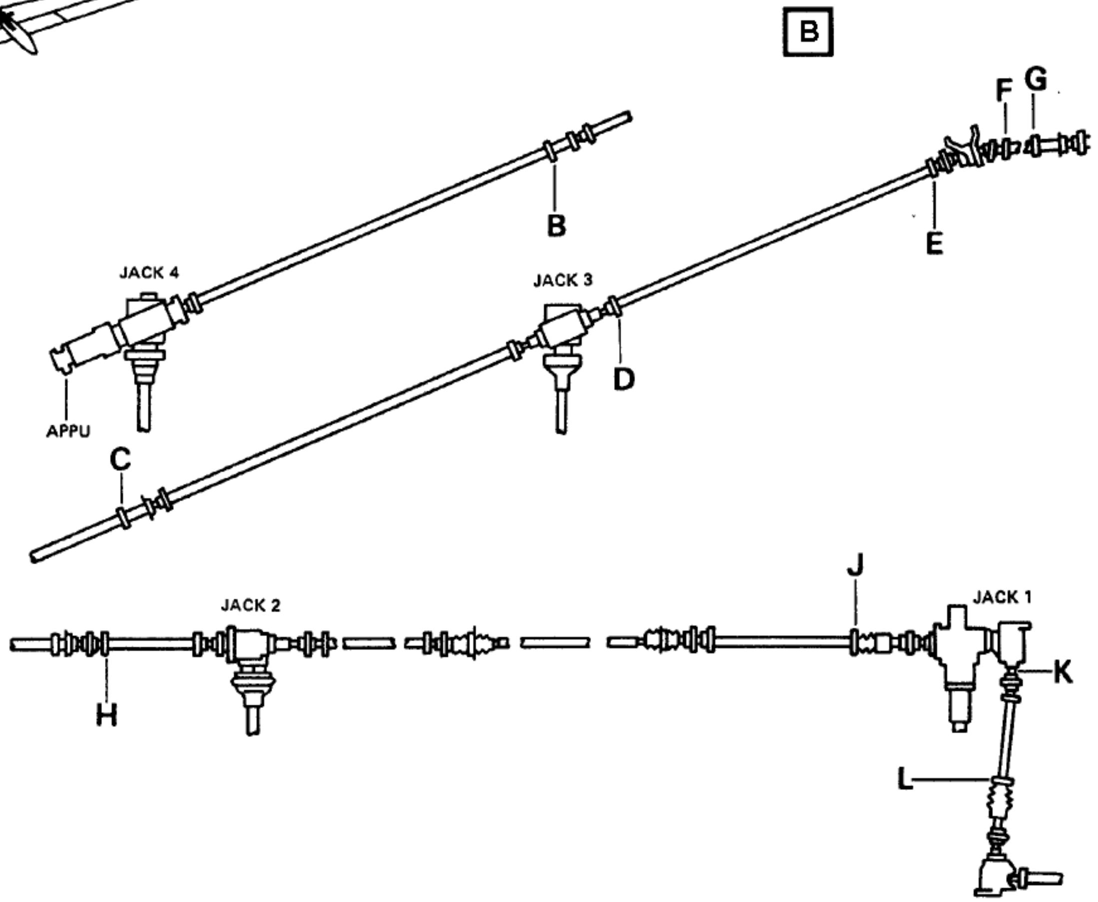
Flaps, slats: power generation and transmission



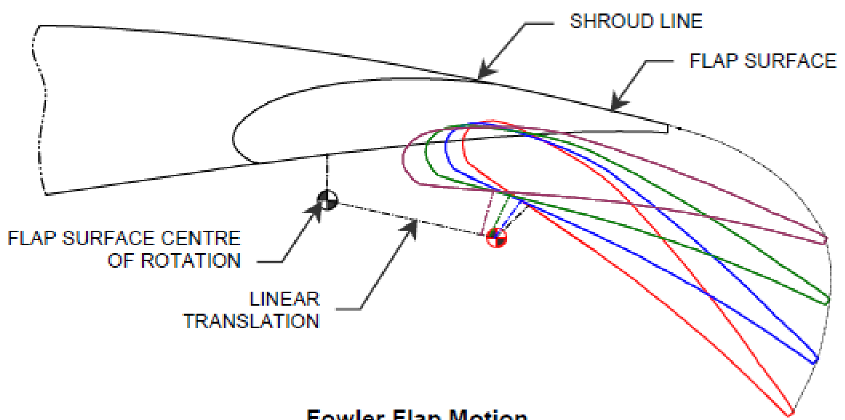
Flaps: mechanical control arrangement



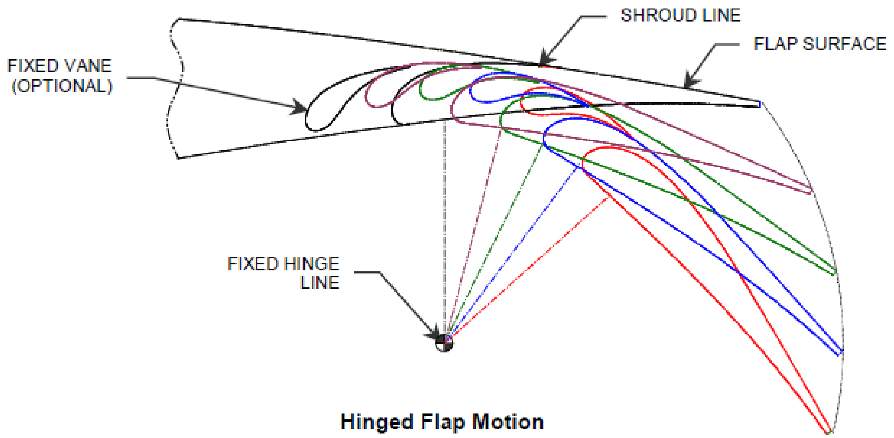
Flap Power Control Unit



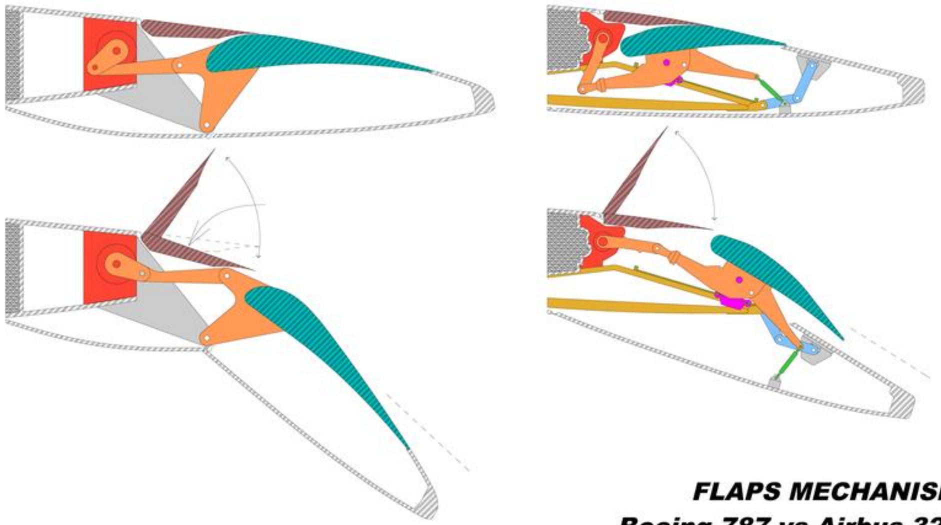
Flaps: control surfaces movements



Fowler Flap Motion

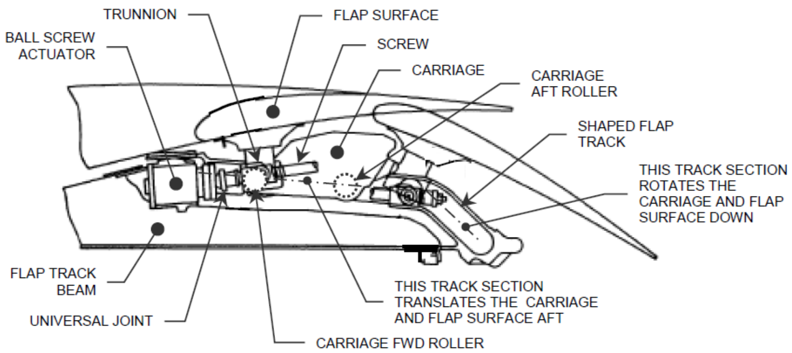


Hinged Flap Motion



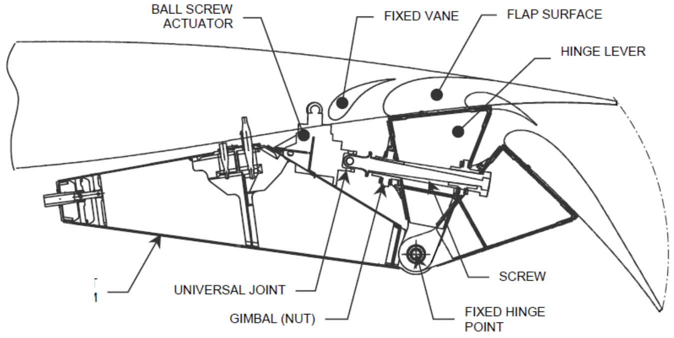
FLAPS MECHANISM
Boeing 787 vs Airbus 320

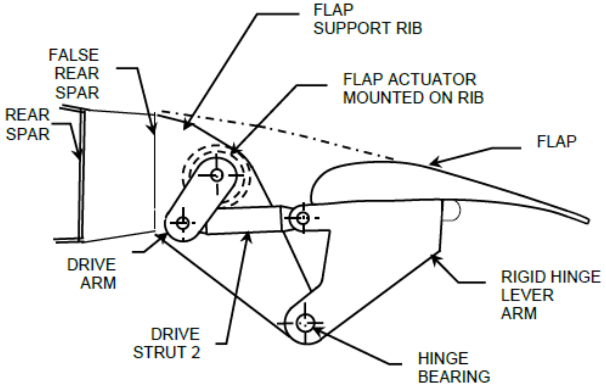
Flaps – Actuation systems



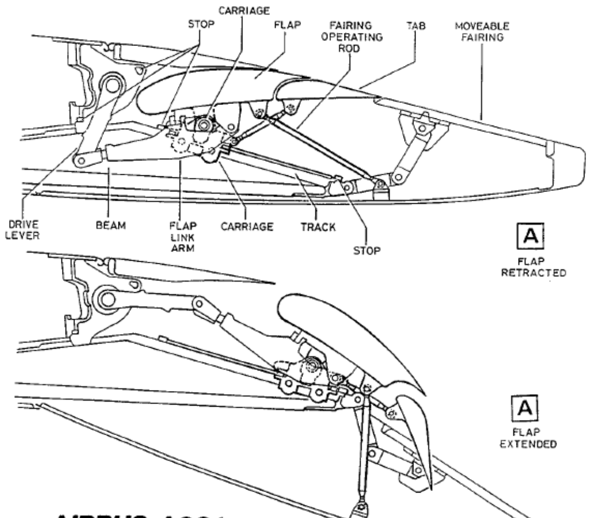
A300/310, linear actuator

A400M, Linear actuator

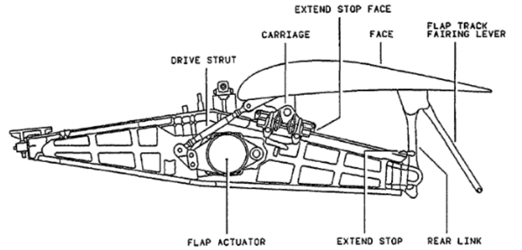




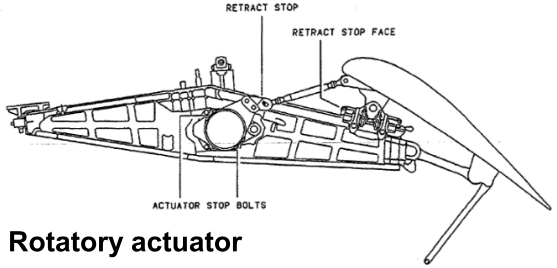
A350, Rotatory actuator



AIRBUS A321
Rotatory actuator

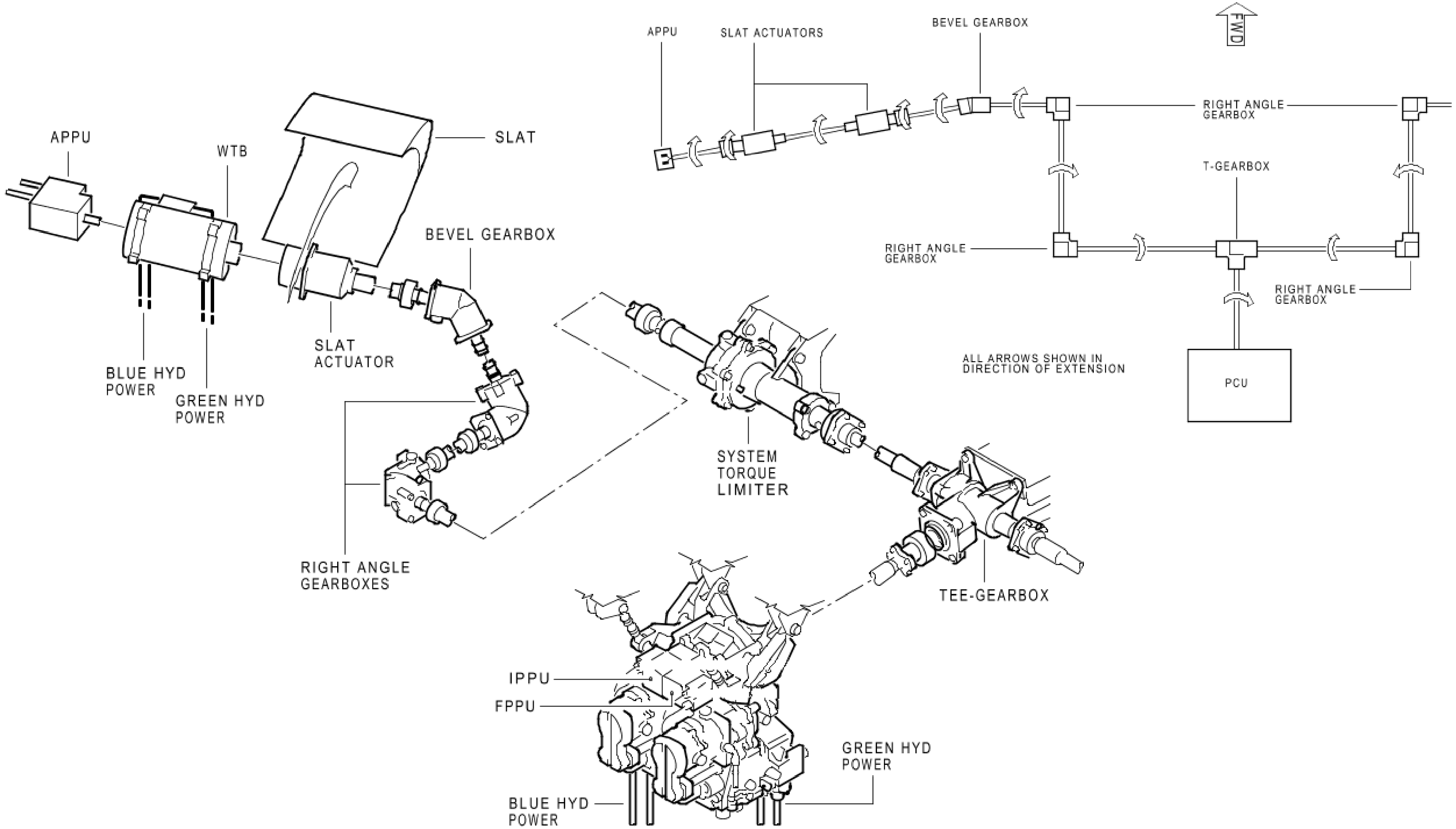


AIRBUS A340-500/600

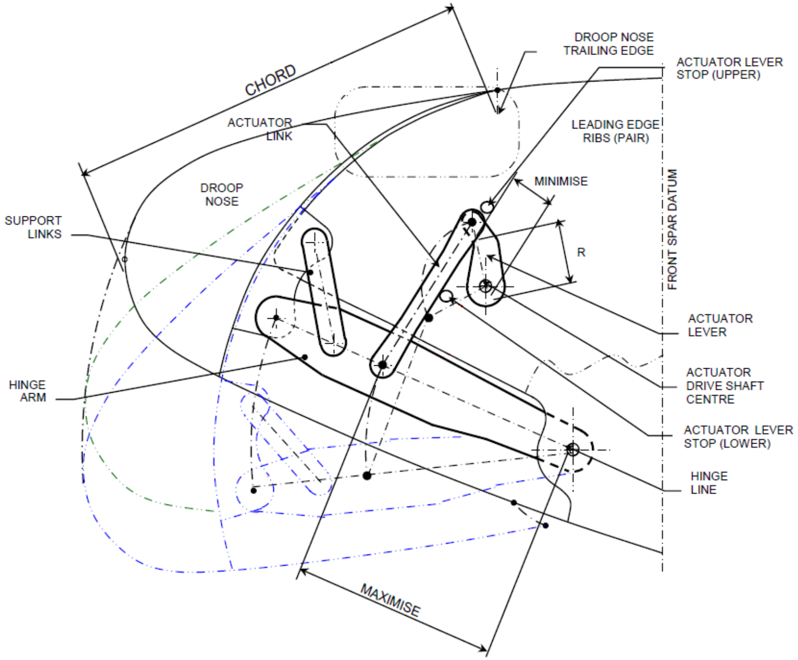


Rotatory actuator

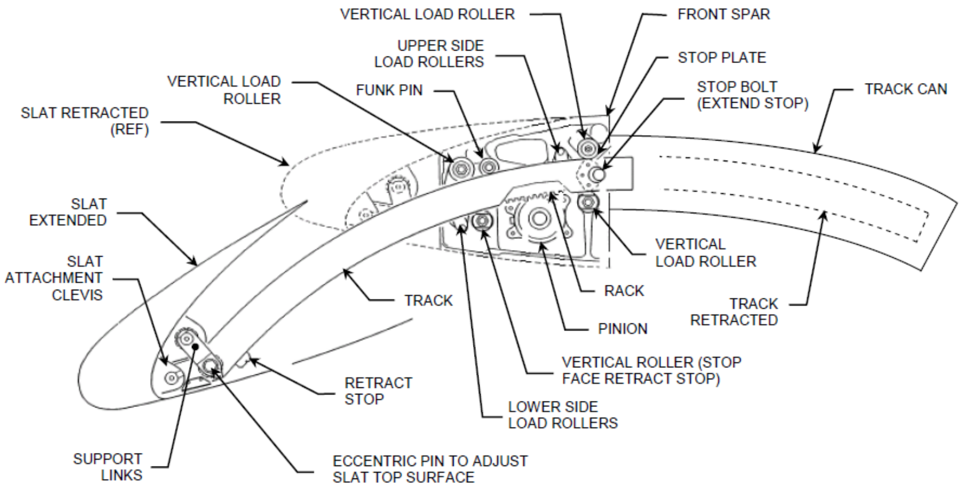
Slats: mechanical control arrangement



Slats: actuation systems



Droop nose system



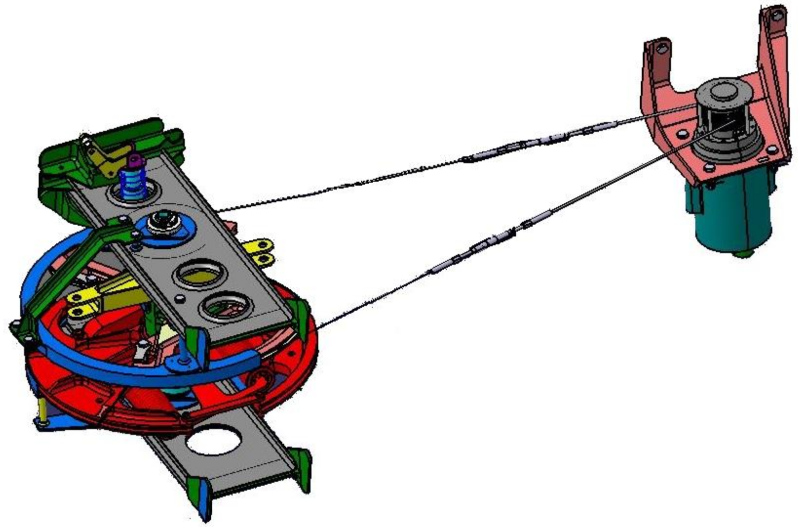
Slat system

Autopilot

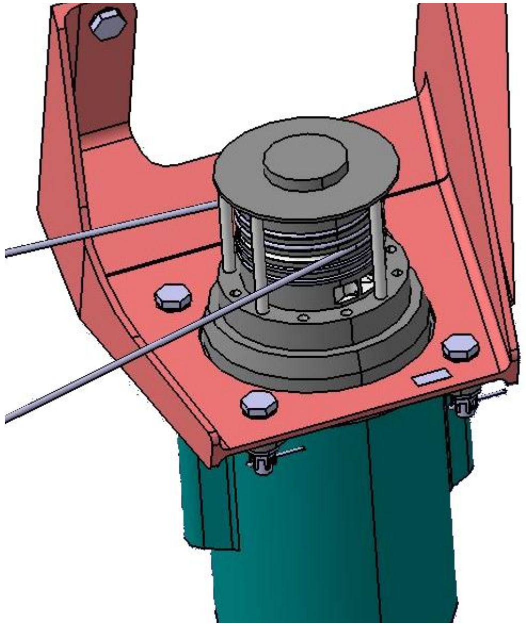
In the case of fly-by-wire installations, the autopilot computer commands the actuators and the position of the pilot controls (stick and pedals) directly.

With the mechanically controlled installations, manual or assisted, it is needed that the autopilot system moves all the mechanical chain.

Similar system is used for the stick-pusher installation.



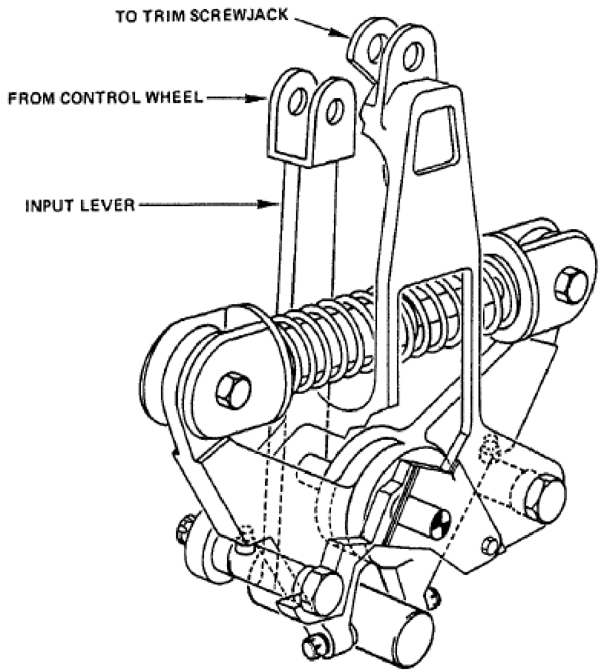
Yaw control - autopilot

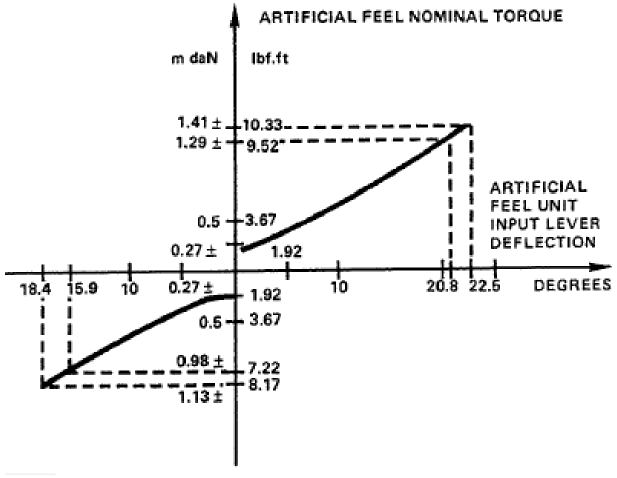


Feel units

When the control surfaces are moved by actuators, artificial feel units are used for:

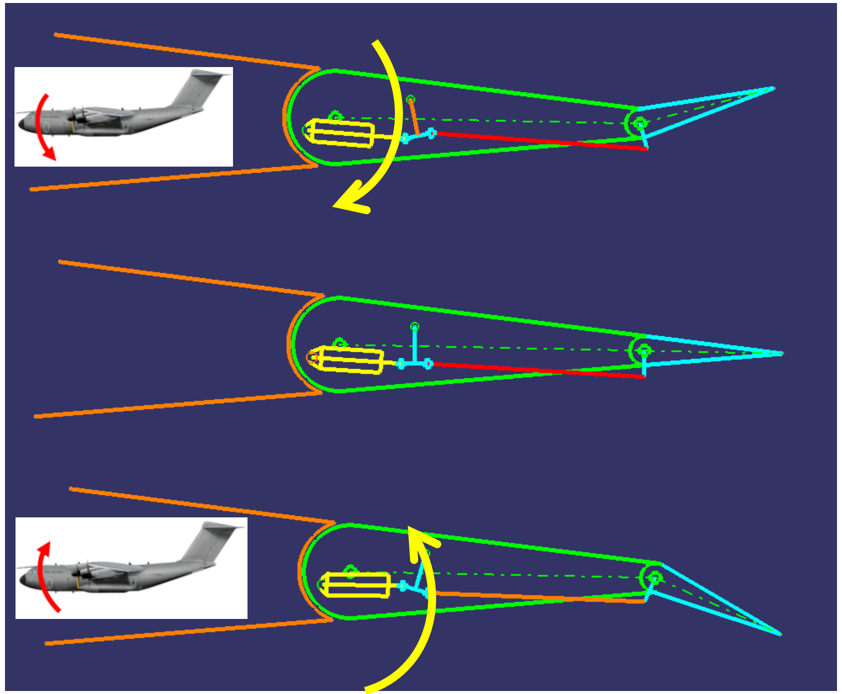
- To provide artificial loads to the pilots proportional to control surface deflection
- To provide accurate centering of the surface at neutral in the absence of a control input



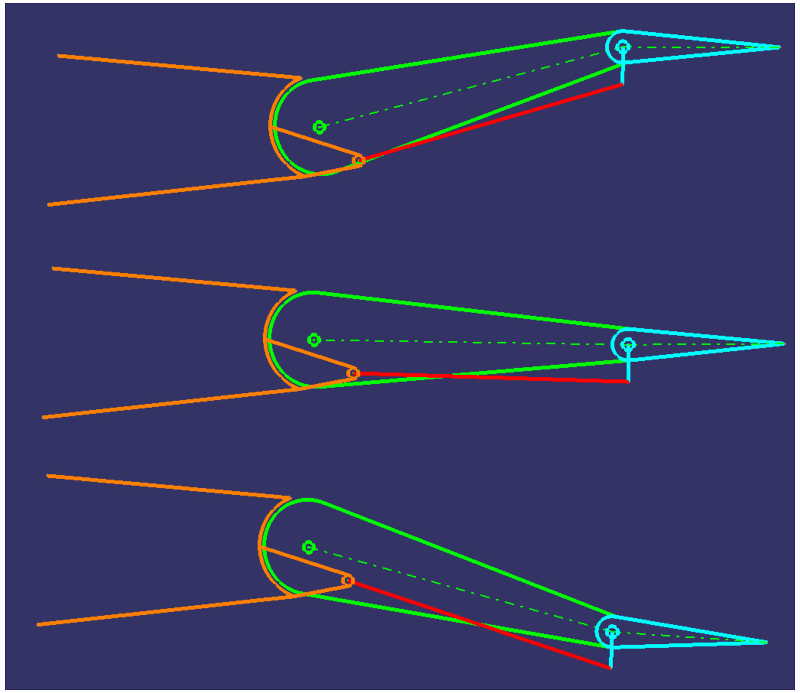


Trim´s & Servos - Manual mechanical installations

- Trims are used to stabilize the aircraft in flight due to CoG changes, cross wind, etc. The tab deflection causes the control surface to rotate in the opposite direction
- Servo compensators are used to alleviated pilot effort to deploy the control surface

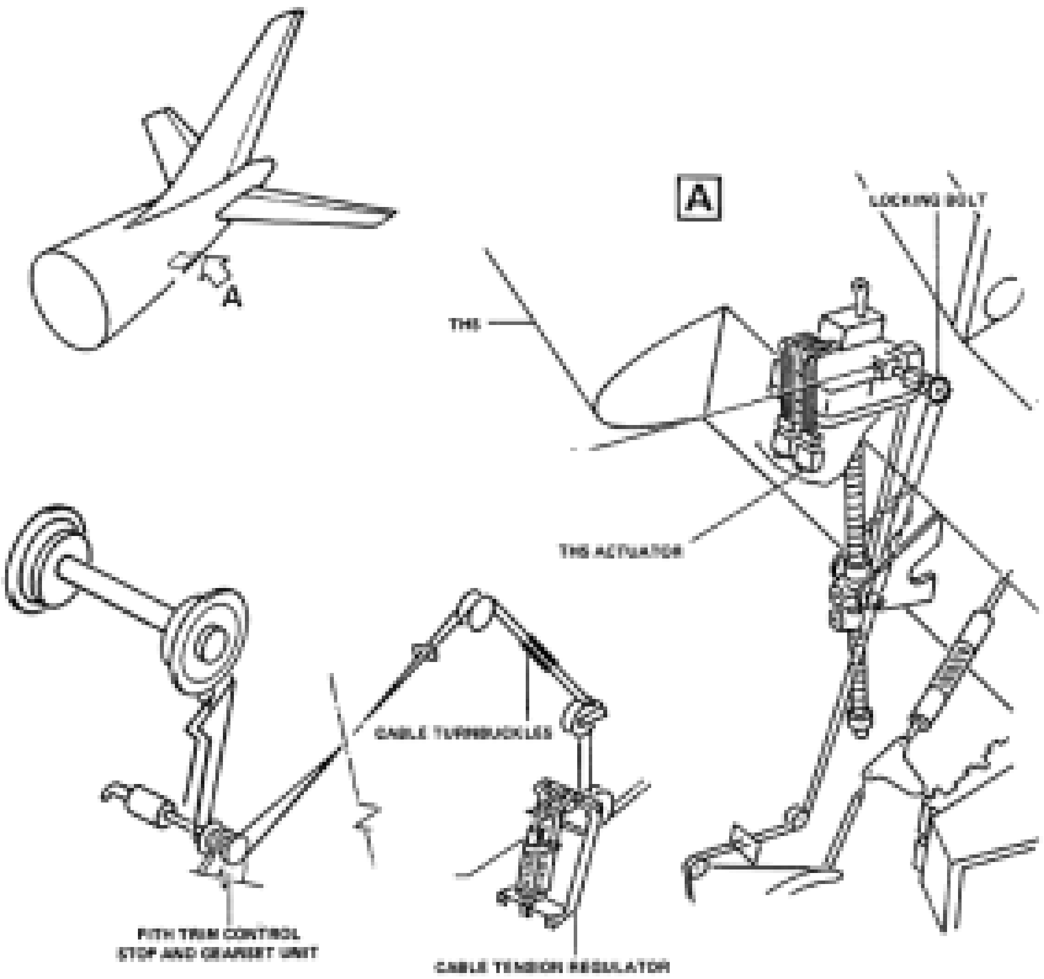


Trim tab



Servo tab

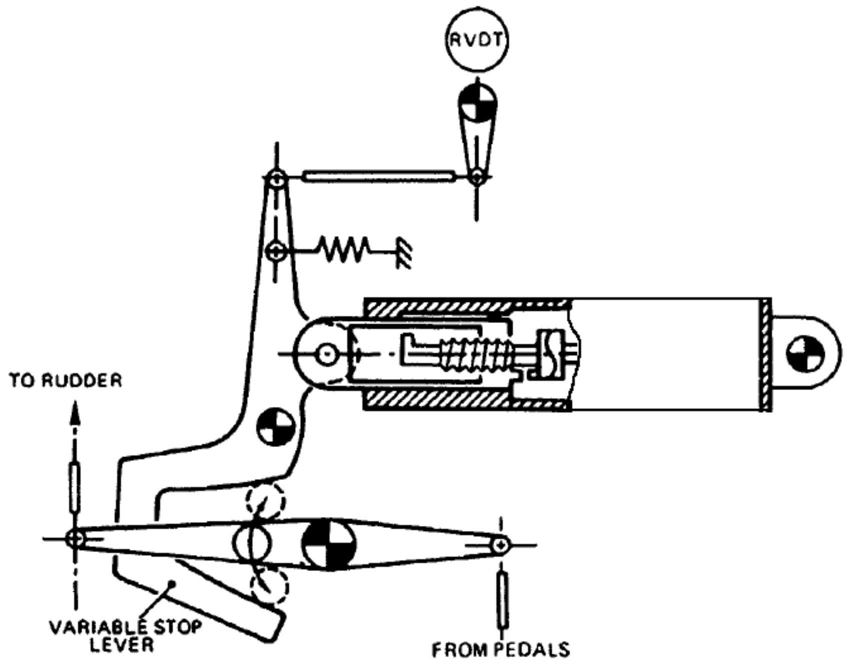
Trims units – Assisted mechanical installations



THS control

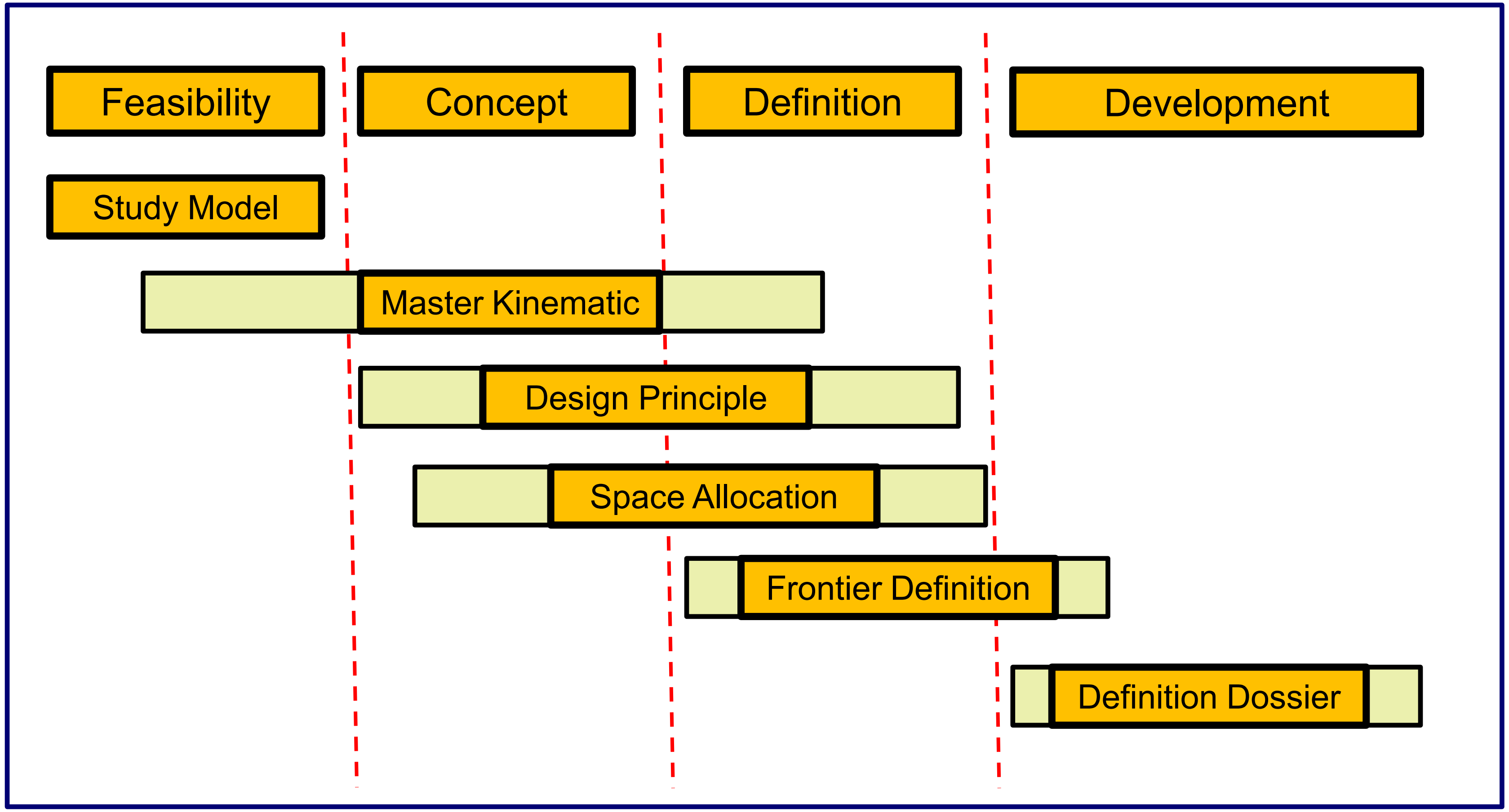
Miscellaneous installations – Control travel limiting system

In some cases is needed to modify the control surface travel in relation to airspeed; limitation is such that the maximum deflection which can be achieved by the control surface remains lower than those which would induce limits loads on the structure throughout the flight envelope



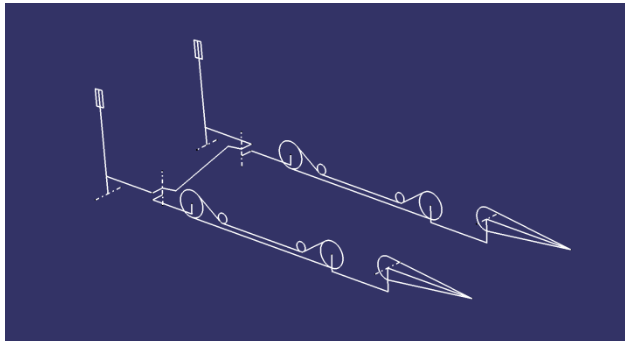
Mechanism Design Process

— The development of a mechanism is an iterative process that involves several disciplines and with different stages



Mechanism Design Process

— Master Kinematics
- The Master Geometry of the Mechanism is called the "Skeleton" (3D Geometry) as all it consists of points, lines, curves and planes. It is also called Master Kinematic (MK).
- The Master Kinematic is designed by adding "joints" between the set of points, lines curves and planes representing the main characteristics/dimensions of all moveable parts of the mechanism.
- The development of a Master Kinematic is an iterative process which, for complex mechanisms, requires to perform various studies allowing to optimize the dimensions/ characteristics of the moveable parts (and which can also have an influence on the design of fixed parts).
- It is generally performed in co-operation between the following domains: Master Geometry, Aerodynamics, Structure, Stress and Systems.
- The analysis of the Master Kinematics is used for:
  → Forces/strokes evaluation
  → Clearance studies
  → Clash detections
  → Digital Mock-Up reviews
  → Human Simulations

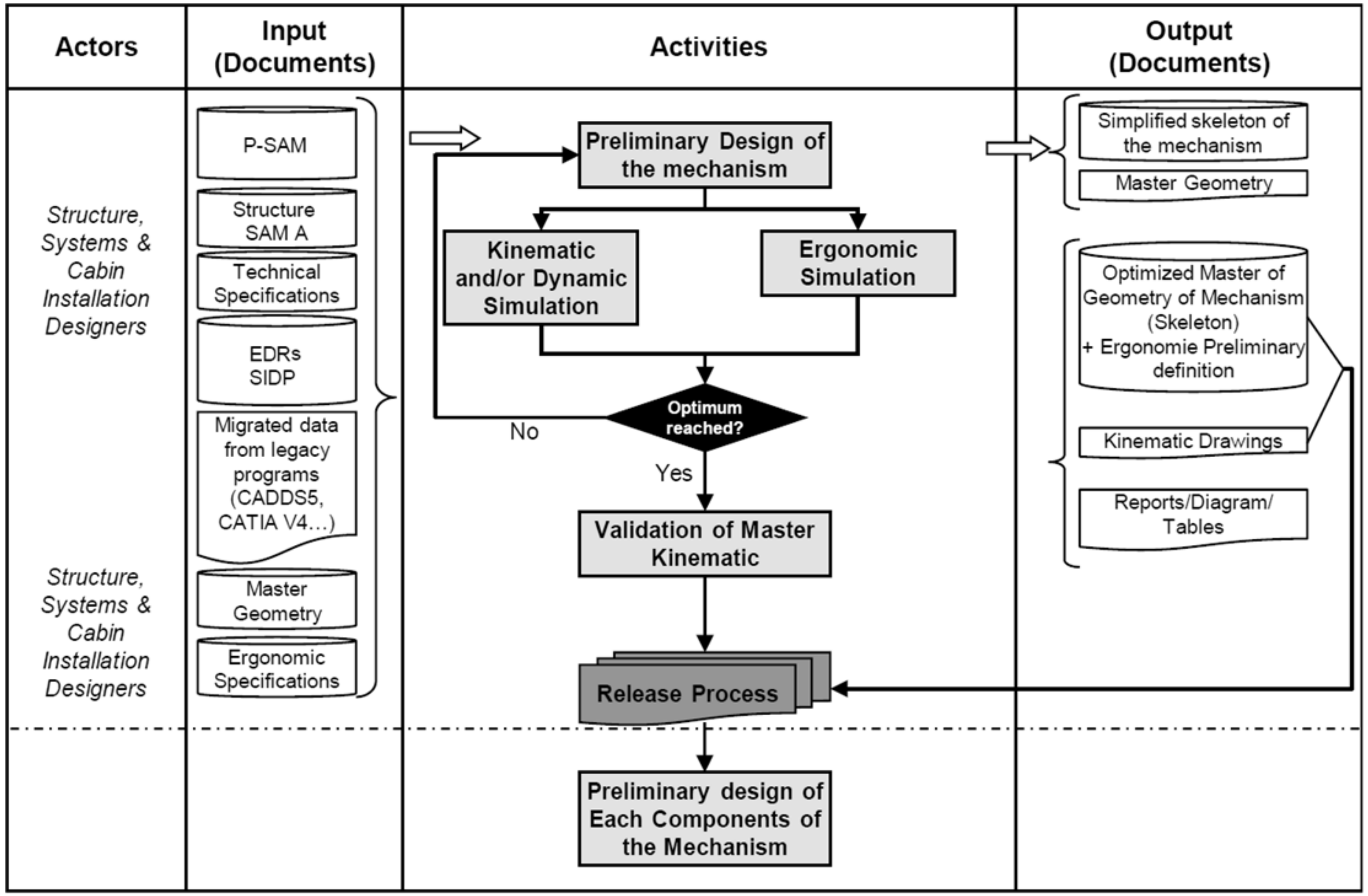


Mechanism Design Process

- Preliminary design of the mechanism
  - The preliminary design of the mechanism is ensured by definition, development and release of a Master Kinematic (MK). The MK is a representation of the main dimensions/characteristics of the moveable parts (and fixed parts).
  - The output of MK process is the released Master Kinematic "Skeleton" and associated Kinematic Drawings or 3D models.
  - During this step, all possible analysis shall be performed to optimize the performances of the mechanism itself
  - Movable parts representations are connected by "kinematic joints" allowing users to simulate and optimize the mechanism.
  - Movement analyses are performed in order to check force, trace, stroke and length

Mechanism Design Process

— Preliminary design of the mechanism

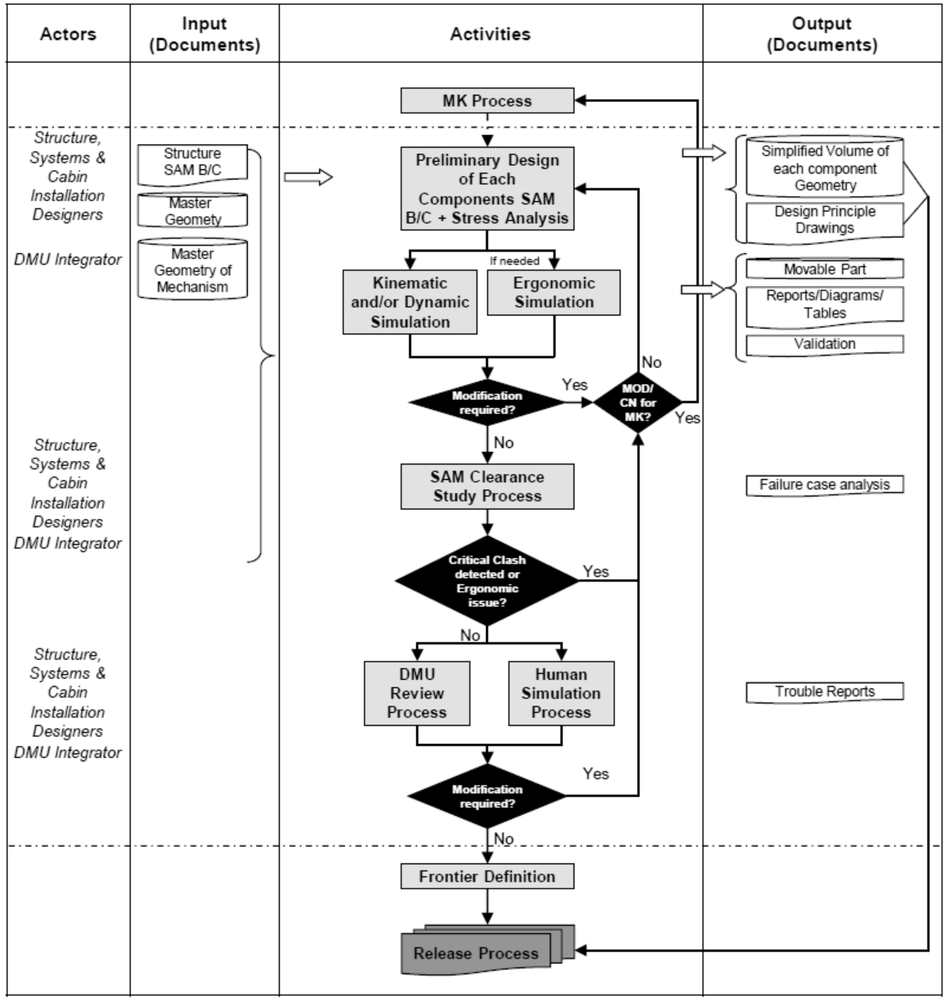


Mechanism Design Process

- Preliminary design of each component/SAM
  - The preliminary design of each component is ensured by creation of preliminary representation of each component of the mechanism (in neutral position). It can be based on wireframe, surface or simplified solid geometry.
  - The output is the released Design Principle. Note that, during preliminary design of each component, it can be needed to modify or make some changes in the Master Kinematic model.
  - At this step, the SAM Parts allow performing additional types of Analysis:
    - → Clearance Study/SAM
    - → DMU Review
    - → Human Simulation

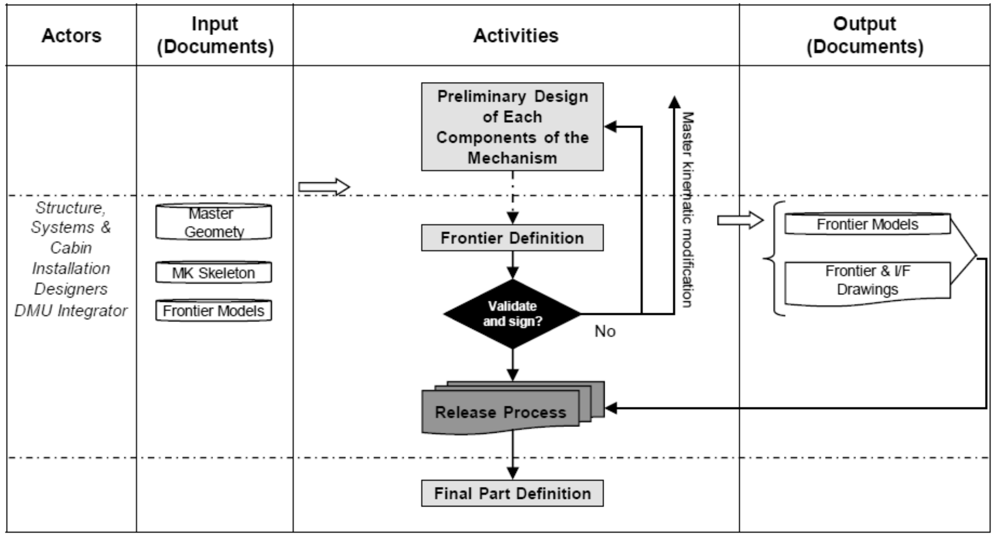
Mechanism Design Process

- Preliminary design of each component/SAM



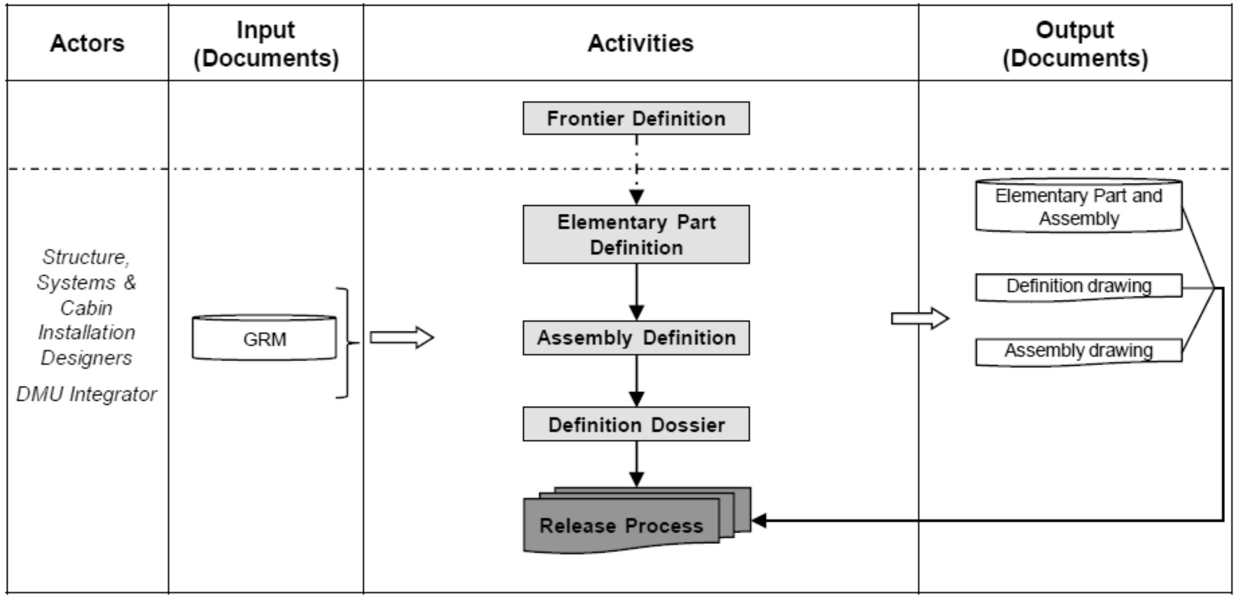
Mechanism Design Process

- Frontier definition
  - In some cases, due to mechanisms complexity, it is needed to define and agree with other design or manufacturing teams on the overall mechanism frontier or interfaces (positions and dimensions)
  - The tolerance analysis of movable parts shall be taken into account and space shall be reserved for:
    - → Over-travels
    - → Whole envelope generated by the movements of movable parts
    - → Interface definition between movable parts and fixed environment
    - → Rigging/adjustment procedure development

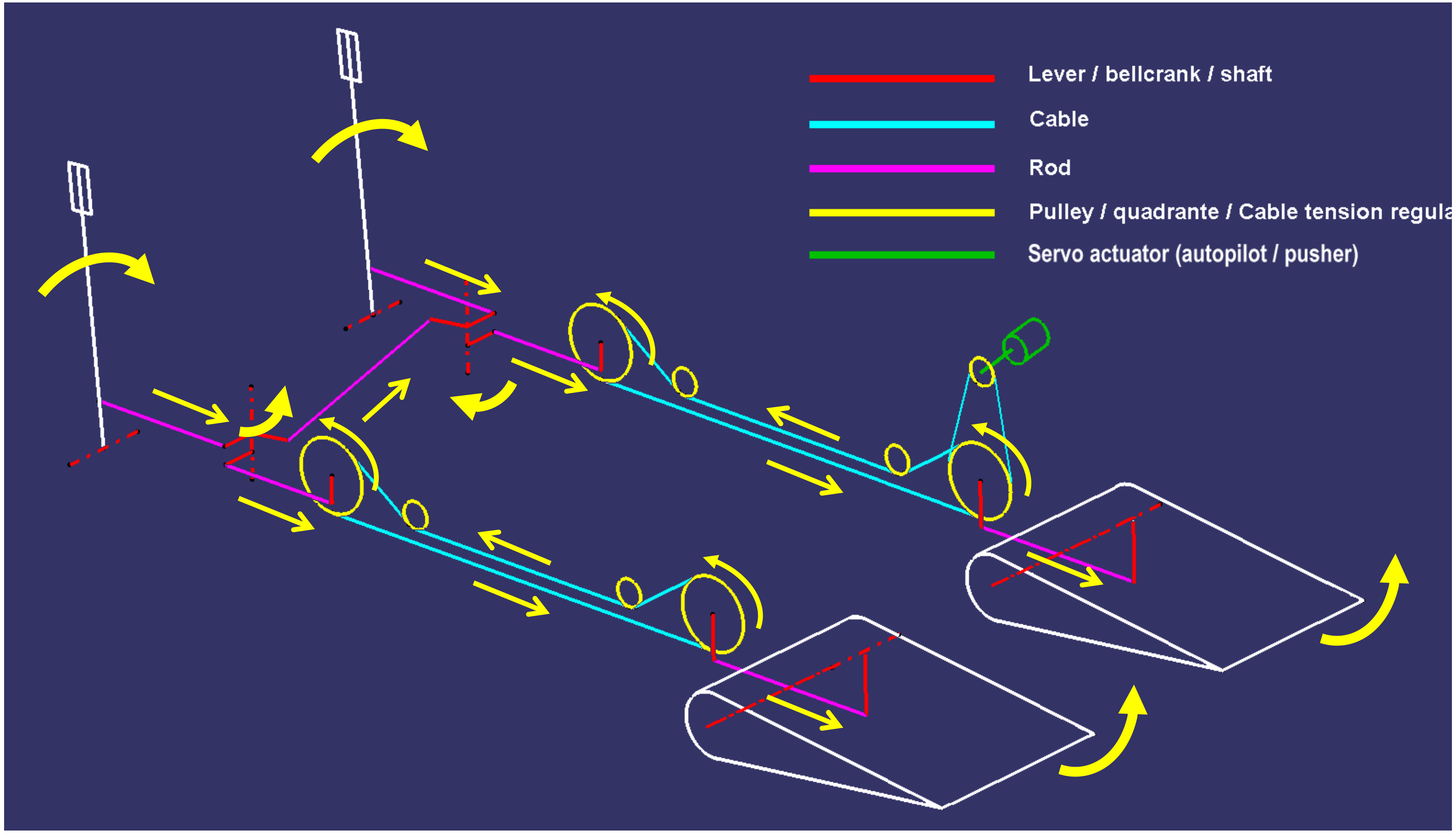


Mechanism Design Process

- Final part definition
  - The Final Part Definition of each component consists in creation of the final 3D Model of all Elementary Parts of the mechanism (in neutral position). All Elementary Parts, Standard Parts and Equipment (if relevant within the mechanism) will be structured in a set of Sub-Assemblies/Assembly representing the definition of the whole mechanism.
  - The Preliminary Design of each component shall be used as a basis for this activity.
  - The output of Final Part Definition Process is the released Design Data Set for all component of the mechanism



Control lay-out



System installation requirements

Routes

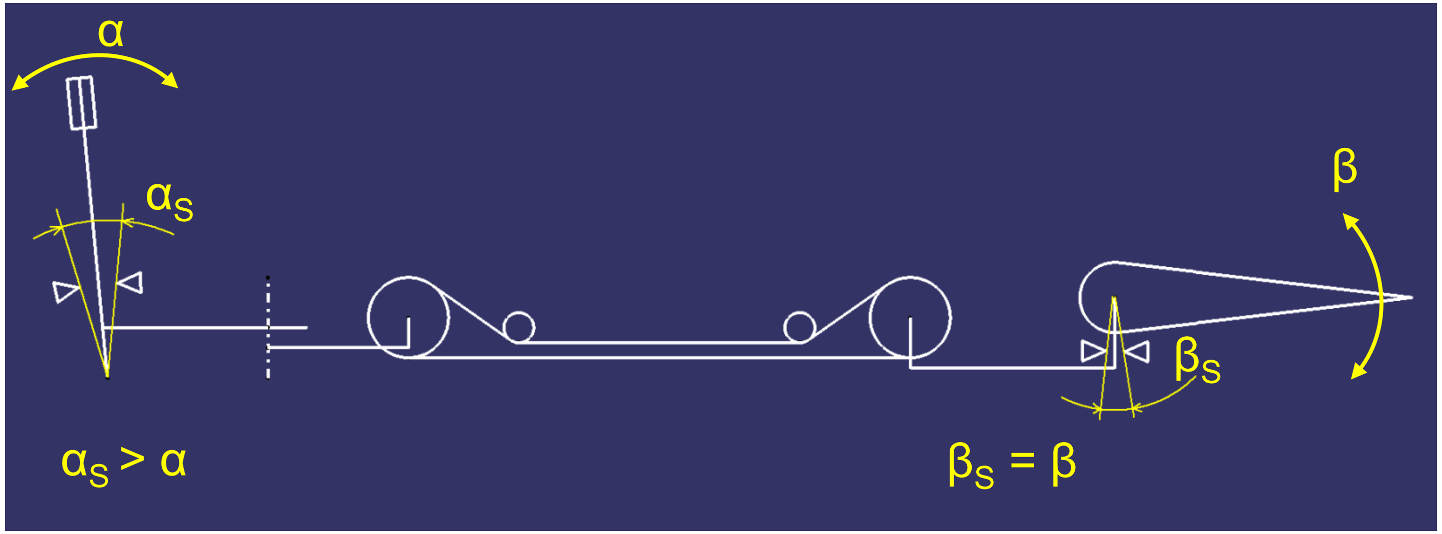
Routing for mechanical controls installations shall be:

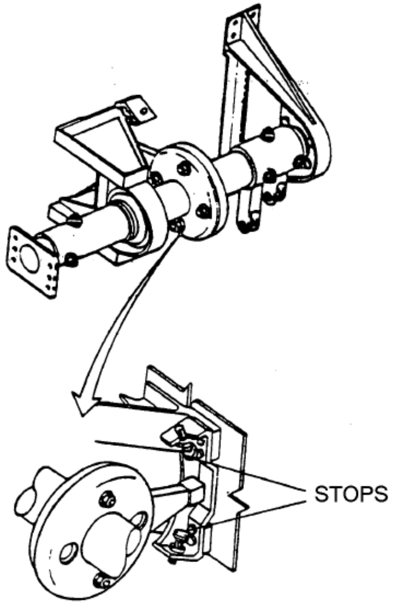
- Short and direct as much as possible
- Systems segregation and duplication
- Hazard areas
  - Engine burst
  - Blade propeller detachment
  - Explosive installations
  - Tire burst
  - Bird strike
- Minimize connections between pressurized and non-pressurized areas
- Ensure easy maintenance access and rigging
  - Installation/removal of adjacent systems must not affect flight controls installations
- Duplicate routing running parallel must be, as much as possible, at the same distance of the aircraft neutral axis and have the same configuration in order to avoid control actuation due to structural deflections or thermal variations

Stops

Pending on the type of actuation, different stops must be provided

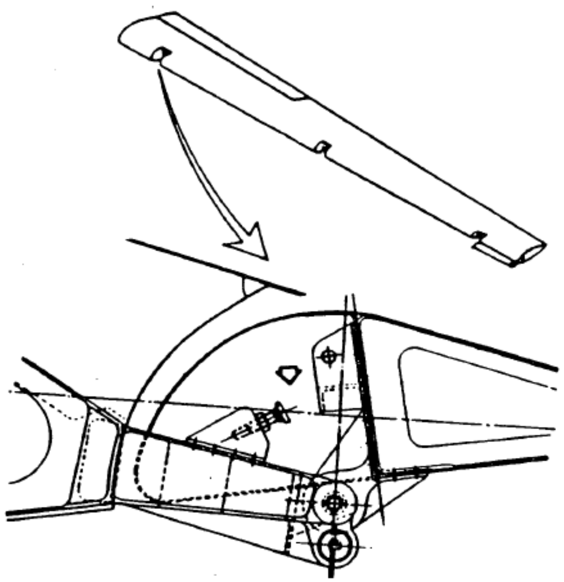
- Mechanical actuation:
  - Nominal deflections: α (control stick) and β (control surface)
  - Input stops (α_S): near the cockpit controls to limit their movements
  - Output stops (β_S): limiting angular deflection of the surface
  - Output stop must be reach before input stops; additional force required to put in contact input stops is called stop cushion force. This gap is needed to absorb elastic deformations of the mechanical control components





Stops

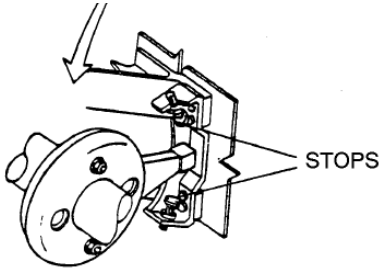
- Assisted/fly-by-wire actuation:
  - Input stops (α): located close to the cockpit area, their main function is, besides to limit the control travel, to protect sensors or actuation valves against overtravels and excessive forces
  - Output stops (β): limiting angular deflection of the surface; if provided internally in the power actuator, additional structural stop should be considered just in case actuator removal (enough clearance must be provided to ensure these stop will not be in contact when the actuators are installed)

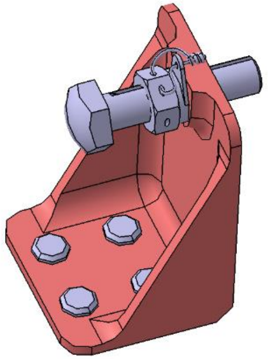


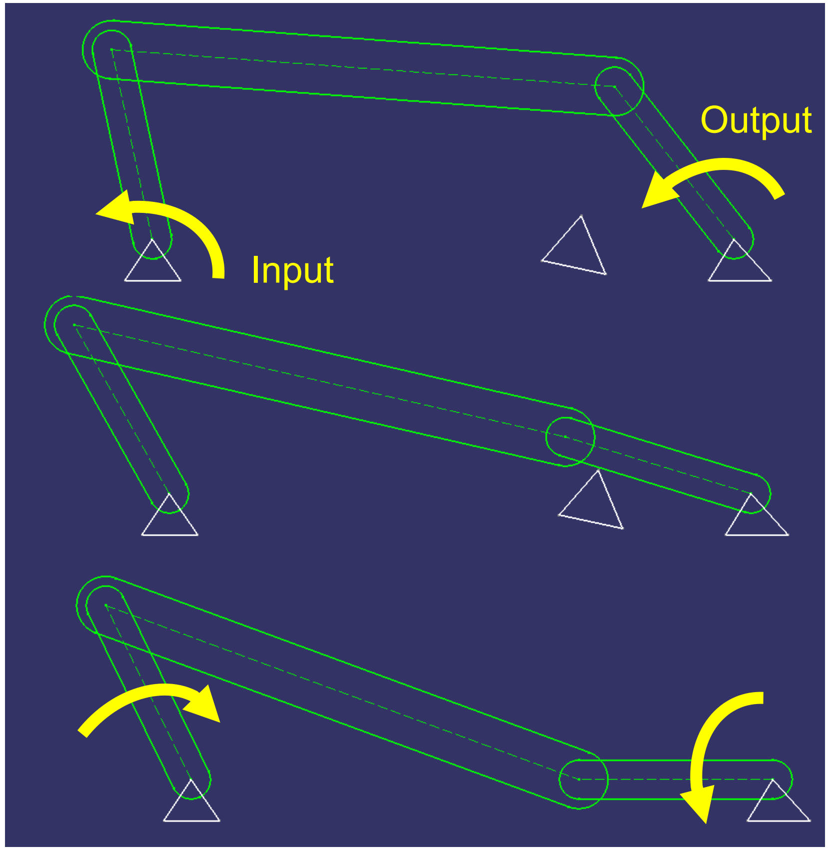
Stops

- Overtravel stops: special precautions must be taken to ensure that the mechanisms con not reach an irreversible position that could lead to a control jamming; in these cases, overtravel stops will be used to avoid the problem

- Adjustable stops must be positively locked in the rigged position







System installation requirements

Rigging

Rigging points shall be minimized as much as possible and it should be possible to replace components, actuators or control surfaces without requiring a complete control system rigging.

Enough space must be provided to perform rigging operations; it is recommended to install bushes in the rigging holes in order to protect the component. Rigging holes which are not bushed shall have enough material provisions to allow a bush to be installed for repair

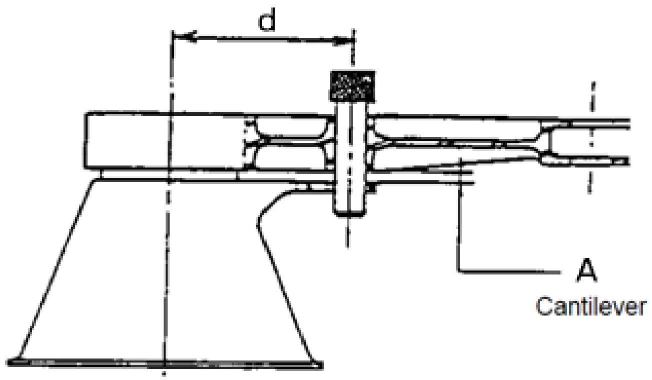
Equipment that requires very precise adjustment operations that need to be performed with specific tools or benches outside the aircraft shall be designed in a such way that the adjustments points are not accessible with the equipment installed

Recommended rigging pins should be according to standards NSA 2020, MS20392-4, for holes in inches, or 8mm diameter pins for metric components.

System installation requirements

Rigging

The pin cantilever length shall be as small as possible considering manufacturing and positioning tolerances and distortion. The distance "d" of rigging hole with respect to rotation axis of the part shall be as large as possible in order to increase the accuracy of rigged position.



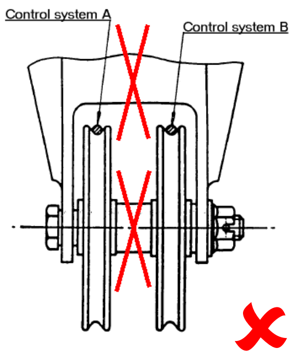
Dim	Ø8H8 mm	Ø .309 - .311 inches
A	2 to 3	0,1 to 0,139

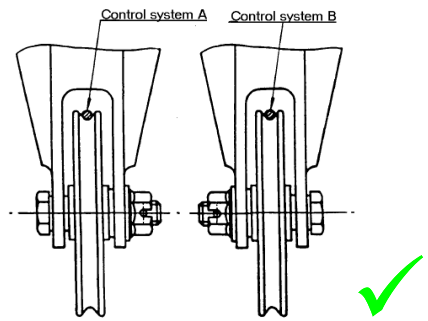
Loose packing blocks or shims should not be used if avoidable, in particular where their omission could cause a strain on the assembly or maladjustment. If possible, shims should be riveted or bonded and properly sealed

Clearance and separation

In case of redundant installations, distance between components of those redundant systems must be maximized as much as possible in order to minimize the possibility of simultaneous failure due to a single issue.

Components of different systems must not share supports or joints; it is needed to segregate control system as much as possible





Enough clearance must be provided for moving parts; for example, for quadrants or rotational components, a minimum overtravel of 5 degrees is recommended.

Next table summarized minimum and recommended clearance for movable parts of control systems

Clearance and separation

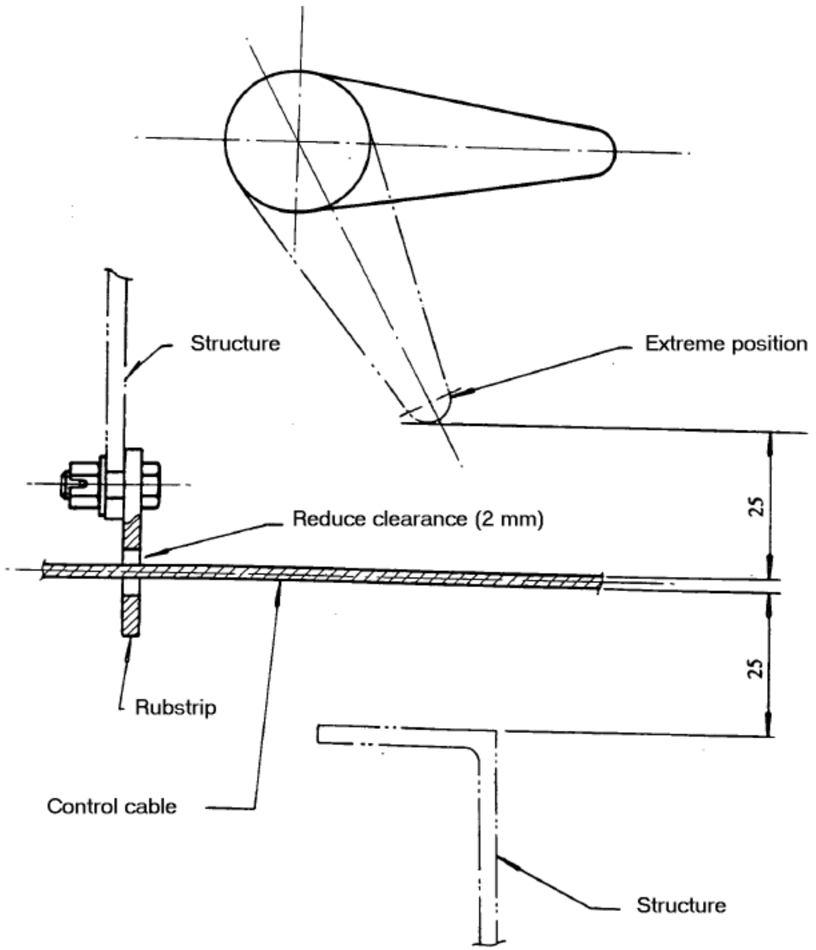
Condition		Clearance between parts 1 and 2	
Part 1	**Part 2**	**Objective**	**Minimum acceptable**
Cable	- Another cable - Structure (no protected) - Structure (protected)	25 mm (1)	15 mm - 25 mm far (i.e more than 300 mm) from a fairlead or a pulley - 13 mm close to fairlead or a pulley 2 mm
Rod	- Structure or fixed equip. - Movable part of another system - Hydraulic hose, electrical loom	25 mm 25 mm 50 mm	13 mm (2) 13 mm (2) 25 mm far from bracket 13 mm near a bracket (2)
Torque shaft (rapid revolut.)	- Structure or fixed equipment		25 mm in the middle between bearings 13 mm near bearings (2)
Bellcrank end or pulley/drum edge	- Fuselage bottom section		25 mm

(1) Turnbuckles coincidence of adjacent control cables should be avoided; in case of coincidence, desirable clearance must be increased up to 38 mm

(2) When manufacturing tolerances are well known, for example along a wing or fin box, a local reduction can be accepted (example : at a rib, at the way through of a rigid pipe...) up to 8 mm if there is no possibility of trapping an object (such as 6.35 mm screw) which may cause jamming

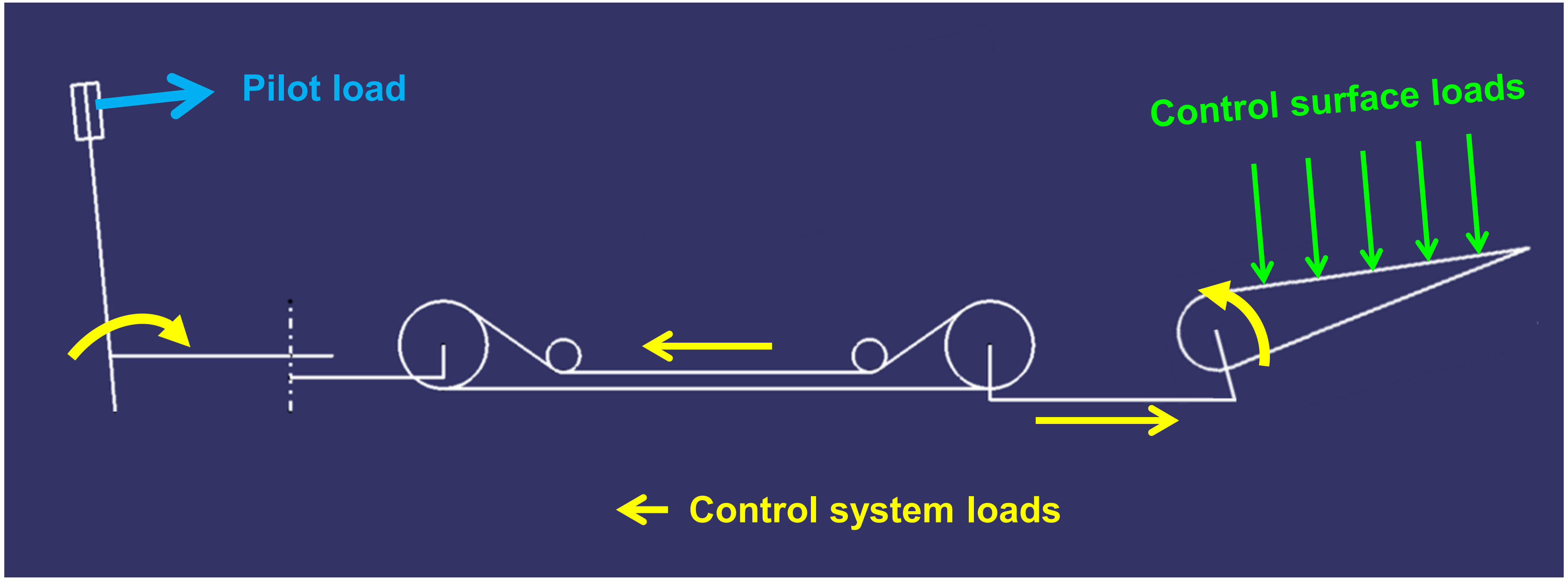
Clearance and separation

- Clearances between parts shall be defined taking into account a loose object (eg: 6,35 mm diameter screw) shall be prevented from being trapped between the movable parts and jamming system operation
- Critical clearances shall be mentioned on the drawing
- Protective covers shall be provided in all critical areas
- It is advisable that complex mechanisms operate in a vertical plane
- Access panels or doors which have to be removed for inspection or maintenance shall not interfere with the full overall freedom and functioning of the controls



Applicable loads

- Loads on control kinematic chain



System installation requirements

Applicable loads

Control surface and system loads are stablished by relevant airworthiness regulations; there are two different kind of loads: design and operational loads

Control system design loads

- CS 25.391 Control surface loads: general

  The control surfaces must be designed for the limit loads resulting from the flight conditions in CS 25.331, CS 25.341(a) and (b), CS 25.349 and CS 25.351, considering the requirements for:
  (a) Loads parallel to hinge line, in CS 25.393;
  (b) Pilot effort effects, in CS 25.397;
  (c) Trim tab effects, in CS 25.407;
  (d) Unsymmetrical loads, in CS 25.427;
  (e) Auxiliary aerodynamic surfaces, in CS25.445.

- CS 25.395 Control system

  (a) Longitudinal, lateral, directional and drag control systems and their supporting structures must be designed for loads corresponding to 125% of the computed hinge moments of the movable control surface in the conditions prescribed in CS 25.391.
  (b) The system limit loads of paragraph (a) need not exceed the loads that can be produced by the pilot (or pilots) and by automatic or power devices operating the controls.
  (c) The loads must not be less than those resulting from application of the minimum forces prescribed in CS 25.397 (c).

Control system design loads

- CS 25.397 Control system loads

The control surfaces must be designed for the limit loads resulting from the flight conditions in CS 25.331, CS 25.341(a) and (b), CS 25.349 and CS 25.351, considering the requirements for:

(a) *General. The maximum and minimum pilot forces, specified in sub-paragraph (c) of this paragraph, are assumed to act at the appropriate control grips or pads (in a manner simulating flight conditions) and to be reacted at the attachment of the control system to the control surface horn.*

(b) *Pilot effort effects.* In the control surface flight loading condition, the air loads on movable surfaces and the corresponding deflections need not exceed those that would result in flight from the application of any pilot force within the ranges specified in subparagraph (c) of this paragraph. Two-thirds of the maximum values specified for the aileron and elevator may be used if control surface hinge moments are based on reliable data. In applying this criterion, the effects of servo mechanisms, tabs, and automatic pilot systems, must be considered.

(c) *Limit pilot forces and torques.* The limit pilot forces and torques are as follows:

Control		Max limit forces or torques	Min limit forces or torques
Aileron:	*Stick* *Wheel*	*445 N (100 lbf)* *356 DNm (80 D in.lb)*	*178 N (40 lbf)* *178 DNm (40 D in.lbf)*
Elevator:	*Stick* *Wheel (symmetrical)* *Wheel(unsymmetrical)*	*1112 N (250 lbf)* *1335N(300 lbf)*	*445 N (100 lbf)* *445 N(100 lbf)* *445 N (100 lbf)*
Rudder		*1335 N (300 lbf)*	*578 N 130 lbf*

System installation requirements

Control system design loads

- CS 25.399 Dual control system

  (a) Each dual control system must be designed for the pilots operating in opposition, using individual pilot forces not less than –
  (1) 0·75 times those obtained under CS 25.395; or
  (2) The minimum forces specified in CS 25.397 (c).
  (b) The control system must be designed for pilot forces applied in the same direction, using individual pilot forces not less than 0·75 times those obtained under CS 25.395.

- CS 25.405 Secondary control system

  Secondary controls, such as wheel brake, spoiler, and tab controls, must be designed for the maximum forces that a pilot is likely to apply to those controls. The following values may be used:

Control	***Max limit forces or torques***
Miscellaneous: Crank, wheel, or lever.	$\left(\frac{25.4+R}{76.2}\right) \times 222N\ (50lbf)$*, but not less than 222 N (50 lbf) nor more than 667 N (150 lbf)* *(R = radius in mm).* *(Applicable to any angle within 20º of plane of control).*
Twist	*15 Nm (133 in.lbf)*
Push-pull	*To be chosen by applicant.*

- Control system ultimate design load =1.5 x Limit load

Control system operational loads

- Maximum operational forces: for the lateral, longitudinal and directional flight controls maximum operational loads are specifies by CS 25.143, paragraph (d)

Force, in Newton (pounds), *applied to the control wheel or rudder pedals*	***Pitch*** *N (lbf)*	***Roll*** *N (lbf)*	***Yaw*** *N (lbf)*
For short term application for pitch and roll control - two hands available for Control	*334 (75)*	*222 (50)*	*---*
For short term application for pitch and roll control – one hand available for Control	*222 (50)*	*111 (25)*	*---*
For short term application for yaw control	*---*	*---*	*667 (150)*
For long term application	*44.5 (10)*	*22 (5)*	*89 (20)*

Secondary control system maximum operating force is not specified, although they should be kept low to avoid pilot overload but, at the same time, high enough to prevent their actuation due to vibrations or accelerations

Control system operational loads

- Brakeout force: this is the minimum force which is needed to initiate the movement of the control surface; it is affected by friction, feel forces, counter balance and centering systems
- Friction must be controlled and minimized in order to decrease the breakout force , to avoid controls centering problems (when a control does not return exactly to its zero position when released) or hysteresis issues.

Reminder: Static strength & stiffness criteria